DAFDet: A Unified Dynamic SAR Target Detection Architecture With Asymptotic Fusion Enhancement and Feature Encoding Decoupling

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***Abstract*— In many military and civilian applications, synthetic aperture radar (SAR) image target detection plays a vital role. However, current methods for SAR target detection generally fail to balance speed and accuracy, thus making it impossible to deploy them to real-world engineering applications. In addition, strong scattering, multiscales, high density, complex background interference, and speckle noise make it remarkably challenging to extract effective target information and disentangle background noise from the target information, ultimately resulting in high missing and false alarm rates. To address these issues, a unified dynamic SAR target detection architecture (DAFDet) with asymptotic fusion enhancement and feature encoding decoupling is proposed in this article. First, a dynamic architecture is constructed by cascading two identical detectors and integrating a designed decision maker. This decision maker can automatically decide the inference route by calculating the difficulty score of an SAR image, which ensures efficient inference speed while achieving high accuracy. Second, an asymptotic fusion enhancement feature pyramid network (AFEFPN) is developed, which can avoid the loss and degradation of target information in multistage transmissions through direct interactions of nonadjacent levels, thereby enhancing the extraction of valid target information. Suppression of background noise is achieved by modeling the importance of different feature channels of the fused features. Finally, a task-oriented decoupled head (TODH) is proposed to boost the localization and classification abilities of the model in complex scenarios. It decouples feature encoding at the source, thus providing task-oriented feature context. Numerous experiments on four widely adopted datasets reveal that DAFDet obtains efficient inference speed and optimal detection accuracy, achieving new state-of-the-art target detection performance. The source code will be provided at https://github.com/yangyahu-1994/DAFDet.**

***Index Terms*— Asymptotic fusion, dynamic neural network, feature encoding decoupling, semantic context, target detection.**

I. Introduction

As AN active microwave imaging sensor, synthetic aperture radar (SAR) works independently of light intensity and weather conditions, and still can provide high-resolution imaging of targets even under harsh conditions. In addition, it features strong penetration for obscured and camouflaged targets. In view of these advantages and the ongoing improvement in SAR imaging technology [1], SAR has been widely applied to environmental monitoring [2], resource investigation, agricultural yield estimation, military reconnaissance [3], and disaster alert. As the foundation of these applications, SAR target detection is being studied extensively and intensively.

Traditional SAR image target detection is mainly dominated by constant false alarm rate (CFAR)-based methods [4], [5], [6], [7], [8], [9], [10]. For example, in order to fully utilize the spatial detail information of high-resolution SAR images, by integrating the intensity of each pixel and the correlations between them into a single feature and analyzing and calculating with that feature, Wang et al. [4] proposed an intensity-space domain CFAR (IS-CFAR) method. It obtains favorable performance, especially for low-contrast target scenes. Since IS-CFAR requires consideration of each pixel and correlations between them, this imposes an extremely high computational burden. Based on this, by means of a joint estimation strategy of thresholds and considering the time stationarity of river clutter, Kuang et al. [5] developed a cell-averaging CFAR (CA-CFAR) detector to detect ship targets within rich river clutter. This method effectively reduces the computation and attains excellent ship target detection performance in broad river clutter scenarios. Furthermore, Hyun and Lee [6] further reduced the time of the whole algorithm by performing only one sorting step as a preprocessing for deciding the CFAR threshold, and finally designed an ordered statistics CFAR (OS-CFAR) detector. Researchers have also recently developed other CFAR-based improved detectors, including TS-LNCFAR [7], OR-CFAR [8], BTS-RCFAR [9],

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and AIS-RCFAR [10]. By precisely modeling the statistical properties of background clutter in SAR images, these CFAR-based detectors exhibit satisfactory detection effects within relatively simple scenes. However, they are generally based on the thought of background clutter statistics, which seriously affects their detection performance [11]. Specifically, the idea requires a strong contrast between the clutter and the target and requires that the statistical characteristics of the background clutter follow a specific distribution. In case of higher resolution and more complex background for SAR images, this may lead to incorrect estimation of the distribution parameters thus increasing the false alarm rate and ultimately resulting in a serious degradation of the detection performance. In addition, they rely severely on hand-designed features, which leads to loss of image information in feature extraction and remarkably poor generalization of the model [12].

In recent years, deep learning (DL) models have been favored by a wide range of researchers for their automatic feature extraction process, excellent feature representation ability, and end-to-end training paradigm, and numerous DL-based SAR image target detection methods [11], [15], [16], [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27] have been proposed. For instance, based on YOLOX [13], Wan et al. [15] developed an anchor-free SAR target detection algorithm by redesigning the backbone, neck, and head of the network employing model lightweight strategy, attention mechanism, and dilated convolution, respectively. This algorithm attains excellent detection accuracy. The way CenterNet [14] locates the center point of a target by key point estimation can effectively avoid missing detections of small targets. On this basis, Cui et al. [11] effectively reduced false alarms resulting from inshore and inland interferences by introducing a spatial shuffle-group enhancement attention module into CenterNet [14] while suppressing some of the noise and extracting stronger semantic features. By elaborately designing an anchor-free framework with skip connections and aggregation nodes, a key-point estimation module, and a channel attention module, Ma et al. [22] proposed an anchor-free SAR ship detection method aimed at fusing multiresolution features, eliminating missing alarms, and reducing false alarms. In addition, by embedding soft thresholding functions into a target detection architecture (TRD) and designing corresponding threshold learning subnetworks, Yang et al. [25] introduced an adaptive denoising enhancement representation learning network, intending to improve the capability of the model to extract effective features under the interferences of natural clutter and speckle noise. Researchers also developed the MKSFF-CNN [26] and MSRIHL-CNN [27] networks by fusing feature information extracted from convolutional kernels of different sizes and different levels of features, respectively, with the aim of enhancing the completeness of target feature representation in SAR images. Through the design of the network architecture, all of these methods provide varying degrees of efficiency or accuracy gains for SAR target detection. However, accuracy improvement often requires the introduction of more complex network structures or modules, while efficiency optimization usually employs model lightweighting strategies [28], implying that these models generally need to be redesigned and trained to achieve possible balances between speed and accuracy, which is extremely inflexible for complex application scenarios. In addition, the feature fusion strategy extensively adopted by these methods creates the problem of degradation and loss of target information due to the natural limitations of multiple transmissions and interactions, damaging the fusion effect of the nonadjacent layers, and ultimately leading to difficulties in extracting effective target information in complex scenes such as strong scattering, complex background interference, and speckle noise. Moreover, the *parameter decoupling* strategy commonly adopted by their detection heads utilizes separate branches to encode task-oriented feature context from the same feature map, resulting in an imbalance between classification and localization in complex scenarios.

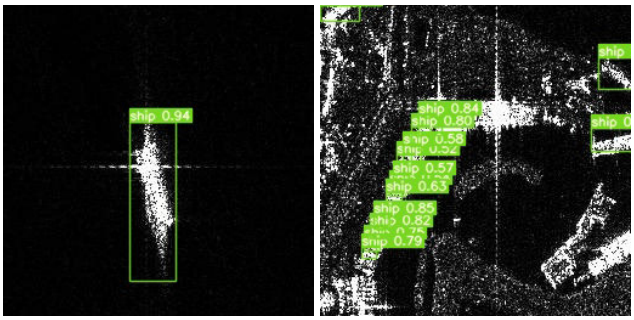


Fig. 1. Representative examples of (a) "easy" and (b) "hard" SAR images. For "easy" and "hard" SAR images, different inference paths are designed, so as to achieve accurate and efficient SAR target detection.

In this article, a unified dynamic SAR TRD (DAFDet) with asymptotic fusion enhancement and feature encoding decoupling is proposed to address these issues. First, the human brain processes images at different speeds, depending on the difficulty of the image [29], [30]. For example, one can accurately and quickly recognize all the targets in Fig. 1(a), while it will take more time to achieve the same effect for Fig. 1(b). Inspired by this, a DAFDet is constructed by cascading two identical detectors and integrating a designed decision maker. This decision maker can automatically decide the inference route by calculating the difficulty score of the SAR image, i.e., for an "easy" SAR image, it only needs to be inferred by the first detector to improve efficiency, while for a "hard" SAR image, it needs to be processed by two detectors to improve accuracy. Which ensures efficient inference speed while achieving high accuracy. In addition, through the adjustment of the threshold in the decision maker, a wide range of speed-accuracy tradeoffs can be achieved with a single model, which consequently satisfies the needs of multiple scenarios. Simultaneously, this dynamic architecture supports the cascading of one- and two-stage detectors. Second, an asymptotic fusion enhancement feature pyramid network (AFEFPN) is developed to effectively extract target information and disentangle background noise from complex scenarios such as strong scattering, multiscales, high density, complex background interference, and speckle noise. AFEFPN is launched through the fusion of two neighboring low-level

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features with varying resolutions and the gradual incorporation of higher level features. Through this asymptotic fusion process, we can greatly shrink the semantic gap between nonadjacent-level features and thus achieve their direct interactions, ultimately avoiding the degradation and loss of the target information in multiple transmissions and interactions. Meanwhile, the adaptive spatial fusion operation mitigates the problem of local information conflict through reasonable spatial location weight assignment thus retaining more valuable fusion information. These guarantee the extraction of effective target information in complex scenarios. In addition, suppression of background noise is achieved by modeling the importance of different feature channels of the fused features. That is, assigning lower or zero weights to the noise feature channels and higher weights to the target feature channels. Finally, by decoupling the feature encoding at the source, a task-oriented decoupled head (TODH) is proposed to boost the localization and classification abilities of the model in complex scenarios. Where, a semantic context and feature sparsity hybrid encoding unit is employed to encode spatially coarser but semantically richer feature maps for the classification task, while a detail and texture encoding unit is utilized to encode high-resolution feature maps containing more edge information for the localization task. In this way, the ever-puzzling coupling between classification and localization is further decoupled.

The main contributions of this article are summarized below.

1) We take the lead in proposing a dynamic architecture for target detection in SAR images. Compared to previous static architectures, this dynamic architecture can meet the latency requirements of multiple scenarios with only adjusting the threshold of the decision maker, and thus is more adaptable to different hardware platforms and changing environments. In addition, the dynamic architecture can be easily adapted to current mainstream one-stage and two-stage detectors.
2) We propose an AFEFPN which can support the direct interactions of nonadjacent-level features, thus avoiding the degradation and loss of feature information in multiple transmissions and interactions, and ultimately enhancing the ability of the model to extract effective target information. In addition, AFEFPN can improve the ability of the model to suppress background noise and disentangle background noise from target information by modeling the importance of different feature channels of the fused features. Furthermore, AFEFPN is highly scalable and can be integrated into most current SAR target detectors.
3) We propose a TODH which further decouples the ever-puzzling coupling between classification and localization by employing different feature encoding units at the source to provide task-oriented semantic contexts for different branches. Compared with the decoupled heads, which generally adopt the *parameter decoupling* strategy, TODH can better improve the ability of the model to classify and localize in complex scenarios. Moreover, it is plug-and-play and can be easily incorporated into most SAR target detection pipelines.
4) Numerous experiments on four widely adopted datasets reveal that DAFDet obtains efficient inference speed and optimal detection accuracy, achieving new state-of-the-art performance.

The remainder of this article is organized as follows. Section II describes the related works in terms of network architecture, feature pyramid, and detection head. Section III provides a detailed description of the proposed methods. In Section IV, the experimental results are presented and analyzed in detail. Finally, the conclusion is given in Section V.

II. Related Works

A. Target Detection Architecture

The development of deep neural networks has given rise to a wide variety of TRDs. Based on the DL network CenterNet [14], Ma et al. [22] designed an anchor-free SAR TRD with aggregation nodes and skip connections in order to fuse multiresolution features and thus detect multiscale ship targets. On the basis of YOLOX [13], Wan et al. [15] proposed an anchor-free TRD, AFSar, by employing model lightweighting, attention mechanism, and dilated convolution to redesign the backbone, neck, and head of the deep network, respectively. By introducing the spatial shuffle-group enhancement attention module into CenterNet [14], Cui et al. [11] proposed an anchor-free TRD SSE-CenterNet, aiming at differentiating targets from background. By optimally fusing high-level deep features with low-level texture and edge features, Ai et al. [27] developed an SAR TRD MSRIHL-CNN for multitarget environments. Apart from these convolutional neural network (CNN)-based architectures, researchers have recently proposed Transformer-based architectures. Given the advantages of CNN in local feature extraction and Transformer in capturing global features, Li et al. [31] proposed a Transformer-based TRD. TRD first adopts a CNN to extract multiscale pyramid features, which are then converted into sequences for input into the Transformer after 2-D positional encoding. Furthermore, by utilizing a shared encoder to efficiently extract multiscale information and three task-specific decoders to obtain different results, Li et al. [32] constructed a Transformer-based TRD, RSCoTr, based on multitask learning. These static TRDs have achieved some success on corresponding tasks. However, they generally inference in a static manner, which greatly limits their representational power, efficiency, and interpretability [33]. Therefore, Liang et al. [34] proposed a novel and flexible network architecture CBNet by grouping multiple identical backbones in a composite connection way. Based on CBNet, Lin et al. [35] proposed a dynamic TRD by cascading two detectors and integrating a designed adaptive router. These dynamic TRDs can adaptively adjust their structures according to different inputs, thus exhibiting remarkable advantages in terms of adaptiveness, accuracy, and computational efficiency.

In this article, we reconstruct the dynamic architecture [35] and based on reconstructed architecture, we propose DAFDet by elaborating the neck and head to achieve the speed-accuracy tradeoffs for SAR target detection in multiple scenarios.

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B. Feature Pyramids

The extraction and fusion of multiscale pyramid features are critical for the target detection task, and feature pyramids are usually employed to effectively extract and fuse multiscale features. To mitigate the effects from strong scattering, by designing a channel spatial enhancement module (CSEM) containing both channel and spatial attention and embedding it at different locations of the path aggregation network, Wan et al. [15] designed a novel feature pyramid (CSEM-PAN). By developing a top-down architecture with lateral connections for constructing high-level semantic feature maps at all scales, Lin et al. [36] proposed the first feature pyramid network (FPN) that exhibits significant improvement in several applications. Based on FPN, Liu et al. [37] developed a path aggregation FPN (PAFPN) by adding a bottom-up information propagation path. PAFPN enables the better fusion of the top-level features with the lower level features thus enhancing the whole feature hierarchy. To improve the efficiency of the model, Tan et al. [38] developed a weighted bi-directional FPN (BiFPN), which allows for easy and fast multiscale feature fusion. By replacing the CSPBlock in the PAFPN with RepBlock or CSPStackRep Block and adjusting the depth and width accordingly, Li et al. [39] innovatively proposed a reparameterization path aggregation network (Rep-PAN). By employing different scale feature maps with different dimensions of channels, Xu et al. [40] proposed RepGFPN, which can better fuse semantic and spatial features of different scales. Aiming at the problems of information degradation and loss in top-down and bottom-up FPNs, Yang et al. [41] proposed an asymptotic FPN (AFPN) for better extraction and fusion of multiscale features. In addition, to eliminate background interference and enhance the ability of the model to represent the target, Gao et al. [42] proposed a hidden recursive FPN (HRFPN) by encoding the difference of information before and after multiscale fusion. HRFPN exhibits promising performance in suppressing background interference and enhancing target discrimination.

Inspired by the insights in AFPN [41], we propose AFEFPN by redesigning the network structure and incorporating the concept of channel enhancement, aiming to enhance the abilities of the model in extracting effective target information and suppressing background noise.

C. Detection Heads

Target detection mainly consists of two subtasks, classification, and localization, which are usually treated employing detection heads. The Coupled Head encodes both classification and localization features with only one 1×1 convolution, and is widely utilized in YOLOv3 [43], YOLOv4 [44], and YOLOv7 [26] due to its efficiency. Considering the conflict between classification and localization tasks, Ge et al. [13] first reduced the channel dimensions utilizing a 1×1 convolutional layer, and then encoded the classification features and localization features, respectively, with two parallel branches, and finally proposed the YOLOX Decoupled Head. For the problem of large spatial information differences of multiscale SAR targets, Wan et al. [15] first designed an ESP module by adopting dilatation convolutions with varying dilatation rates, and replaced the 1×1 convolution in YOLOX Decoupled Head with this module thus proposing the ESP-Head. By adopting a hybrid-channel strategy, Li et al. [39] constructed an Efficient Decoupled Head that achieves a lower inference latency. By employing different modules to encode task-specific semantic contexts, Zhuang et al. [45] further disentangled the feature encoding for classification and localization tasks and proposed a novel task-specific context decoupling (TSCODE) head. Furthermore, by utilizing a partial channel fusion state module and a multidirectional distance attention module to provide the required quality representation for classification and localization, respectively, Gao et al. [42] proposed a semi-coupling task-balanced (SCTB) head. SCTB provides information with different focuses for classification and localization subtasks, achieving consistent alignment between the two tasks.

Inspired by TSCODE [45], we propose TODH by designing separate feature encoding modules at the source and depth-wise separable convolutions at corresponding branches, which provides task-oriented semantic contexts while maintaining lightweight.

III. Proposed Method

This section first elaborates on the proposed DAFDet, including its dynamic processing flow, the design of the decision maker, the training strategy, and the variable-speed inference strategy. Its key components, namely, the AFEFPN and the TODH, are then described sequentially.

A. Dynamic SAR TRD

Fig. 2 illustrates the proposed DAFDet. The developed dynamic architecture consists mainly of two identical detectors and a decision maker. Specifically, for an input SAR image $\mathbf{X} \in \mathbb{R}^{H\times W\times C}$, it is first fed into the Backbone 1 (for convenience, abbreviated as B_1) to extract multiscale features F_1 as

$$F_1 = B_1(\mathbf{X}) = \left[F_1^1, F_1^2, \ldots, F_1^L\right] \tag{1}$$

where H, W, and C denote the height, width, and number of channels of the input SAR image $\mathbf{X}$, respectively, and L indicates the stage number. This multiscale features F_1 is then fed to the decision maker Ψ to calculate the difficulty score $\Phi \in (0, 1)$ for this SAR image as

$$\Phi = \Psi(F_1). \tag{2}$$

According to the calculations of the decision maker, if the SAR image is classified as an "easy" image, it will exit directly at B_1, i.e., the AFEFPN N_1 and the TODH D_1 will output the detection results $\widehat{\mathbf{Y}_1}$ as

$$\widehat{\mathbf{Y}_1} = D_1\left(N_1\left(F_1'\right)\right) \tag{3}$$

$$F_1' = \left[F_1^1, F_1^2, \ldots, F_1^{L'}\right]\left(L' \le L\right). \tag{4}$$

Conversely, if the decision maker classifies the SAR image as "hard," the multiscale features F_1 need to be further processed rather than decoding F_1' directly by N_1 and D_1 and giving the

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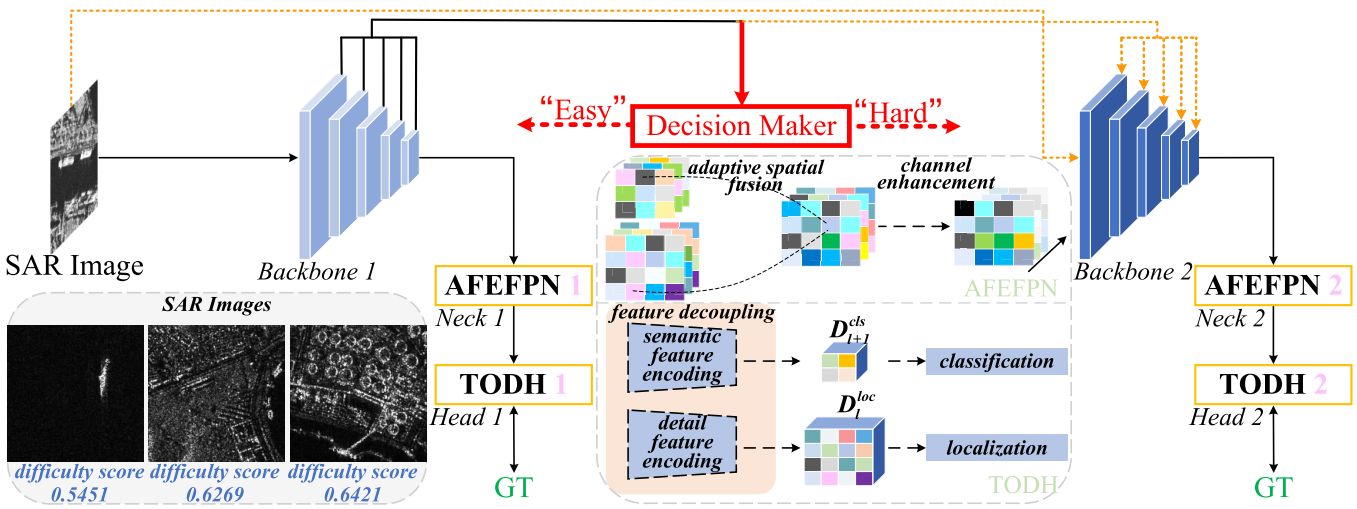


Fig. 2. Proposed DAFDet. Backbone 1 first extracts multiscale features F_1 from the input SAR image. F_1 is then sent to the decision maker to calculate the difficulty score of that SAR image to determine the inference path. If it is an "easy" image, F_1' is decoded directly by N_1 and D_1 and the detection results are output more efficiently. Conversely, in the case of a "hard" image, F_1 will be mapped to G and different scale features with the same number of channels in $g^{\{i\}}$ will be fused at each stage of B_2 in turn to enhance the feature representation of B_2. Similarly, F_2' is decoded by N_2 and D_2 to obtain more accurate detection results.

detection results. Specifically, we first map F_1 into G via a dense higher level composition (DHLC) [30] operation as

$$G = \mathrm{DHLC}(F_1) = \left[g^{\{1\}}, g^{\{2\}}, \ldots, g^{\{k\}}\right] \tag{5}$$

where $g^{\{i\}}$ $(i = 1, 2, \ldots, k)$ denotes a tuple composed of i-scale features. Simultaneously, the input SAR image $\mathbf{X}$ is fed into Backbone 2. Different scale features with the same number of channels in $g^{\{i\}}$ are then sequentially fused at each stage of B_2 to enhance the feature representation of B_2, which can be expressed as follows:

$$F_2 = B_2(\mathbf{X}, G) = \left[F_2^1, F_2^2, \ldots, F_2^L\right] \tag{6}$$

finally, the detection results can be attained by

$$\widehat{\mathbf{Y}_2} = D_2\left(N_2\left(F_2'\right)\right) \tag{7}$$

$$F_2' = \left[F_2^1, F_2^2, \ldots, F_2^{L'}\right]\left(L' \le L\right). \tag{8}$$

As part of the dynamic architecture, the design of the decision maker must take full account of multiscale information, as features at different scales together determine the difficulty score of the SAR image to be judged. Therefore, inspired by SENet [46], we first squeeze the global spatial information of the features at each scale in F_1 into channel descriptors by global average pooling (GAP) and then concatenate them along the channel dimension, denoted as

$$\widetilde{F}_1 = \mathrm{Concat}\left(\left[\mathrm{GAP}\left(F_1^1\right), \mathrm{GAP}\left(F_1^2\right), \ldots, \mathrm{GAP}\left(F_1^L\right)\right]\right) \tag{9}$$

in this way, we squeeze the multiscale features F_1 into a d-dimensional vector $\widetilde{F}_1 \in \mathbb{R}^d$, where d is the sum of the number of channels for each scale feature. Next, we map $\widetilde{F}_1$ to the difficulty score Φ via two fully connected layers, expressed as

$$\Phi = \sigma\left(\mathbf{W}_2\left(\delta\left(\mathbf{W}_1\widetilde{F}_1 + \mathbf{b}_1\right)\right) + b_2\right). \tag{10}$$

Of which, δ and σ indicate the ReLU and Sigmoid activation functions, respectively. $\mathbf{W}_1$, $\mathbf{b}_1$, $\mathbf{W}_2$, and b_2 are the parameters to be fit in the decision maker. Unlike SENet [46], we limit the number of nodes in the first fully connected layer to $\lfloor d/4 \rfloor$ to reduce the computational burden. Furthermore, we set the number of nodes in the second fully connected layer to 1.

As for the training of the dynamic architecture, it differs slightly from most static SAR TRDs. First, two cascades of identical detectors are trained jointly as

$$\min_{\Theta_1, \Theta_2}\left(\mathcal{L}_{\mathrm{det}}^1(\mathbf{X}, \mathbf{Y} \mid \Theta_1) + \mathcal{L}_{\mathrm{det}}^2(\mathbf{X}, \mathbf{Y} \mid \Theta_2)\right) \tag{11}$$

where Θ_i indicates the learnable parameters of detector i and $\mathcal{L}_{\mathrm{det}}^i$ indicates its training loss, i.e., the sum of the classification loss and the regression loss. $\mathbf{X}$ and $\mathbf{Y}$ indicate the input SAR image and ground truth (GT), respectively. After finishing the above training, these two detectors will be able to detect the target as normal. Next, we freeze the parameters Θ_1 and Θ_2 of detectors 1, 2 and train the decision maker to automatically distinguish the difficulty of the SAR image. The ideal situation is that the decision maker can assign "easy" SAR images (i.e., with lower Φ) to a faster detector (i.e., detector 1) and "hard" ones (i.e., with higher Φ) to a more accurate detector (i.e., detector 2). Nevertheless, it is extremely challenging to achieve this in practice. Therefore, we redefine the difficulty criterion for SAR images, i.e., an SAR image should be identified as an "easy" SAR image if the training loss difference between the two detectors of the SAR image is sufficiently small, and as a "hard" one if it is sufficiently large. Based on this definition, we expect a state of equilibrium to exist when training the decision maker, where the easier half of the SAR images in the training set are processed by detector 1 and the harder half are processed by detector 2. Inspired by deep reinforcement learning [47], we achieve a dynamic balance by introducing a dynamic offset to balance the losses of detectors 1 and 2. In this case, the training objective of the decision maker can be expressed as

$$\min_{\Theta_d}\left((1-\Phi)\left(\mathcal{L}_{\mathrm{det}}^1(\mathbf{X}, \mathbf{Y} \mid \Theta_1) - \Delta/2\right) + \Phi\left(\mathcal{L}_{\mathrm{det}}^2(\mathbf{X}, \mathbf{Y} \mid \Theta_2) + \Delta/2\right)\right) \tag{12}$$

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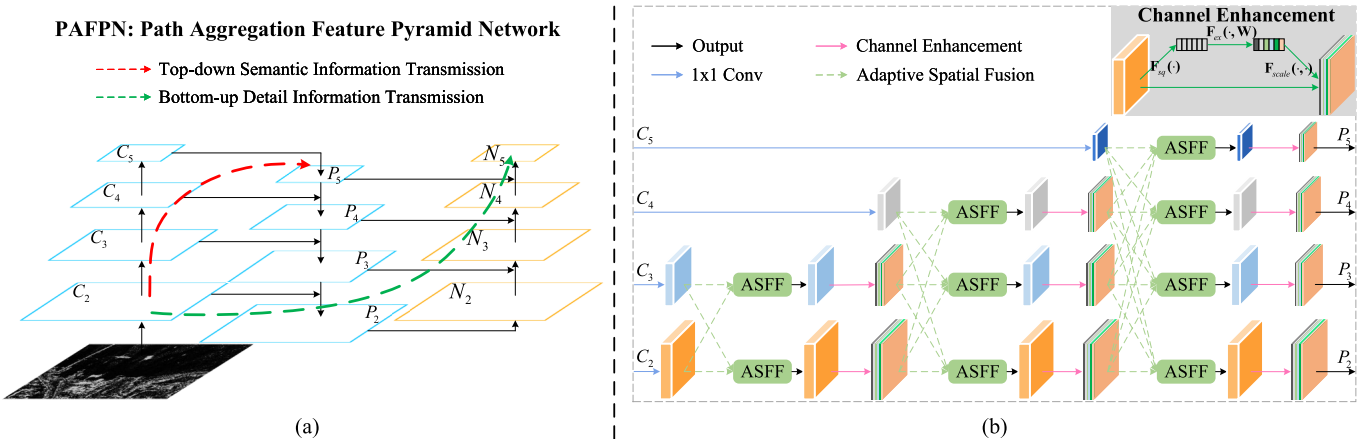


Fig. 3. Architecture diagrams of (a) PAFPN [37] and (b) proposed AFEFPN. First, a set of features $\{C_2, C_3, C_4, C_5\}$ with different scales are extracted from the backbone bottom-up. In this bottom-up procedure, different levels of features C_2, C_3, C_4, and C_5 are fused asymptotically. Finally, the channel-enhanced features $\{P_2, P_3, P_4, P_5\}$ with four different scales are directly output to detection heads.

where Θ_d denotes the learnable parameters of the decision maker. Δ denotes the median of the training loss difference between detectors 1 and 2 on the training set. $\Delta/2$ indicates the dynamic offset, which is applied to reward and punish detectors 1 and 2, respectively. Thus, the decision maker can be optimized by introducing the gradient of the difficulty score Φ to the parameter Θ_d as

$$\frac{\partial \mathcal{L}}{\partial \Theta_d} = \frac{\partial \mathcal{L}}{\partial \Phi}\frac{\partial \Phi}{\partial \Theta_d} = -\frac{\partial \Phi}{\partial \Theta_d}\left(\mathcal{L}_{\text{det}}^1 - \mathcal{L}_{\text{det}}^2 - \Delta\right). \tag{13}$$

It is clear that the optimization direction of the decision maker is closely related to the difficulty of the SAR image.

In the inference phase, difficulty score thresholds need to be determined to ultimately achieve speed-accuracy tradeoffs under different scenarios with only one DAFDet. Inspired by the method for determining the overlap threshold when training a linear SVM classifier in R-CNN [48], they can be determined in this way. First, the difficulty score set $\mathcal{S}_{\text{val}}$ of the SAR images on the validation set is calculated. The corresponding threshold is then calculated based on the scene constraint (i.e., the target latency) as

$$\eta = \frac{\text{lat}_t - \text{lat}_1}{\text{lat}_2 - \text{lat}_1}, \quad \text{lat}_1 \le \text{lat}_t \le \text{lat}_2 \tag{14}$$

$$\eta' = (1 - \eta) \times 100\% \tag{15}$$

$$\tau_{\text{val}} = \text{percentile}(\mathcal{S}_{\text{val}}, \eta') \tag{16}$$

where lat_t is the target latency associated with a particular scene. lat_1 and lat_2 denote the latencies of detectors 1 and 2, respectively. η indicates the maximum proportion of SAR images classified as "hard." percentile$(\mathcal{S}_{\text{val}}, \eta')$ indicates calculating the η'th quantile of $\mathcal{S}_{\text{val}}$. Since the validation and test sets are independent and identically distributed, τ_{val} can be generalized to the test set, that is,

$$\tau_{\text{test}} = \tau_{\text{val}}. \tag{17}$$

B. Asymptotic Fusion Enhancement FPN

The architecture of the proposed AFEFPN is shown in Fig. 3(b). As with conventional FPN- or PAFPN-based [see Fig. 3(a)] target detection methods, four scales of features are extracted from the last layer of the different feature stages of the backbone before entering the neck for feature fusion, denoted as $\{C_2, C_3, C_4, C_5\}$. In this bottom-up feature extraction process, AFEFPN asymptotically enhances the fusion of the low-level, high-level, and top-level features, and finally, outputs channel-enhanced fusion features $\{P_2, P_3, P_4, P_5\}$, with steps of $\{4, 8, 16, 32\}$. In particular, the low-level features C_2 and C_3 are initially fed into the AFEFPN, and a 1×1 convolution is employed to compress the number of channels so as to reduce the parameters and calculations. The adaptively spatial feature fusion is then carried out by means of the ASFF [49]. Prior to this, feature resizing is mandatory. For up-sampling, we apply a transposed convolution based on the desired up-sampling rate to adjust the channel and resolution simultaneously instead of a 1×1 convolution for aligning the channel and then a bilinear interpolation for up-scaling the resolution as in [49]. This leads to less information redundancy and stronger mapping capability. Similarly, we implement down-sampling employing convolutions with different convolution kernel sizes and strides according to the required down-sampling rate. Suppose $x_{ij}^{n \to l}$ indicates the feature vector at position (i, j) on the feature map resized from layer n to layer l. Performing an adaptive spatial fusion of multiscale features on level l, we have

$$y_{ij}^l = \alpha_{ij}^l \cdot x_{ij}^{1 \to l} + \beta_{ij}^l \cdot x_{ij}^{2 \to l} \tag{18}$$

where y_{ij}^l indicates the vector at position (i, j) of the fused feature map y^l. α_{ij}^l and β_{ij}^l denote the spatial weights at position (i, j) of feature maps $x^{1 \to l}$ and $x^{2 \to l}$ resized to layer l, follow the constraint $\alpha_{ij}^l + \beta_{ij}^l = 1$ and are adaptively learned by the network. In this way, we assign different spatial weights to features at different levels during multilevel feature fusion, enhancing the importance of critical levels and mitigating the impact of conflicting information from different targets. However, noting that α_{ij}^l and β_{ij}^l are scalar variables and shared across all channels, instead of designing multiple cascaded residual units [50] following each feature fusion, we employ a single SENet [46] for channel enhancement. This is motivated

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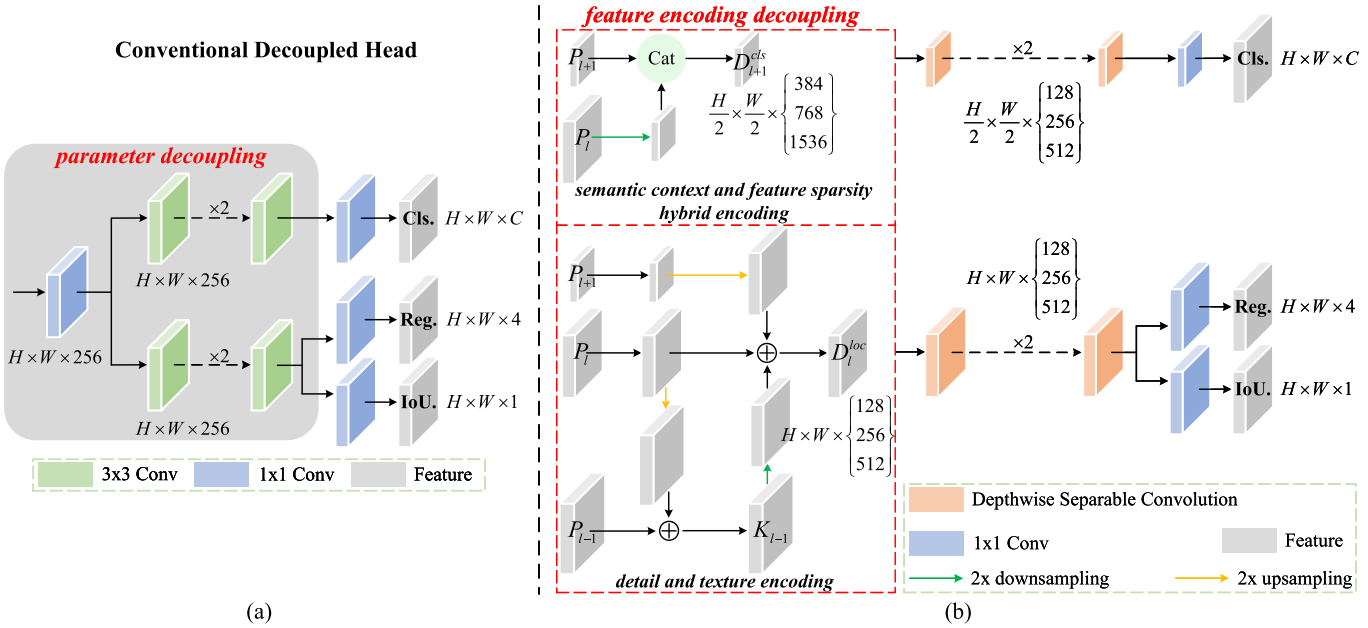


Fig. 4. Illustrations of (a) conventional decoupled head [13], [39] and (b) proposed TODH. The developed TODH first decouples the feature encoding at the source thus offering task-oriented feature maps for the classification and regression branches. The task-oriented feature maps D_{l+1}^{cls} and D_l^{loc} are subsequently projected by depthwise separable convolutions and decoded by 1×1 convolutions to obtain the final predictions.

by the fact that adaptive spatial fusion essentially establishes the importance of different regions of space, but ignores the importance of different feature channels, while the immediately following channel enhancement operation compensates for it by modeling the importance of the individual feature channels and thus improves the representational ability of the model for features.

Next, high-level features C_4 and top-level features C_5 are sequentially fed into the AFEFPN until adaptive spatial fusion and immediately following channel enhancement are completed for all scales. Finally, we output the channel-enhanced fusion features $\{P_2, P_3, P_4, P_5\}$ with four different scales directly to detection heads. It is worth noting that we neither repeat the 1×1 convolution after compressing the number of channels with 1×1 convolution, nor do we use 1×1 convolution to roll back the number of channels after channel enhancement. In addition, the semantic difference between nonadjacent-level features is larger than that between adjacent-level features, especially for low-level features C_2 and top-level features C_5. This directly leads to poor fusion of nonadjacent-level features. Since the fusion process of AFEFPN is gradual, this will bring the semantic information of disparate feature levels closer together, thus alleviating the above problem.

C. Task-Oriented Decoupled Head

Following the neck of the network, the output multiscale feature maps are leveraged for classification and regression predictions. Unlike typical decoupled heads [see Fig. 4(a)] that feed the classification and regression branches with a certain single-scale same feature map, the TODH receives inputs from three scales and decouples the feature encoding at the source thus providing task-oriented features for both branches, as illustrated in Fig. 4(b). Specifically, for classification task, it is relatively coarse-grained and relies on rich semantic context. Traditional methods of determining classification results by critical or salient parts of a target result in severe feature redundancy in the classification branch, as these important components may be sparsely distributed. Furthermore, higher abstraction and understanding of richer semantic contexts help in target classification. For example, ships are more likely to be present at the same time as ships than with airplanes, and occluded ships can be recognized by a larger view. In addition, the downsampling operation has almost no impact on classification performance and significantly reduces the computational burden. These in-depth insights motivate us to design a semantic context and feature sparsity hybrid encoding module for accurate and efficient classification. As shown in Fig. 4(b), this module first performs twofold downsampling of the feature map P_l at pyramid level l employing a 3×3 convolution with a stride of 2 to reduce the computation. Concatenation with the semantically contextually richer feature map P_{l+1} is then carried out at level $l+1$ to generate classification-oriented features D_{l+1}^{cls}

$$D_{l+1}^{\text{cls}} = \text{Concat}(\text{DConv}(P_l), P_{l+1}) \tag{19}$$

where $\text{Concat}(\cdot)$ and $\text{DConv}(\cdot)$ denote the concatenation operation and downsampling convolution, respectively. Next, we take the two depthwise separable convolutions as a feature projection function and act on D_{l+1}^{cls}. Each depthwise seperable convolution consists of two parts: a 3×3 depthwise convolution and a 1×1 pointwise convolution. The advantage of this design is that it can significantly reduce the number of parameters and computational cost, achieve separation of channels and regions, and dramatically boost efficiency without sacrificing model performance. Finally, the projected features are decoded into classification scores with a 1×1

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TABLE I
QUANTITATIVE COMPARISON OF FOUR WIDELY ADOPTED SAR TARGET DETECTION DATASETS

Dataset	Category	Sensor	Polarization	Size	No. of samples
MSAR-1.0 [53]	bridge ship airplane oil tank	Gaofen-3 HISEA-1	Single	256∼2048	28449
SSDD [52]	ship	TerraSAR-X Sentinel-1 RadarSat-2	Single	190∼668	1160
LS-SSDD-v1.0 [55]	ship	Sentinel-1	Single	24000×16000	15
SAR-Ship-Dataset [54]	ship	Sentinel-1 Gaofen-3	Single Dual Full	256×256	43819

convolution. Since the resolution of D_{l+1}^{cls} is $(H/2) \times (W/2)$, each position (x, y) in D_{l+1}^{cls} needs to predict the classification scores of four categories, i.e., classification prediction $\tilde{C} \in \mathbb{R}^{H_{l+1} \times W_{l+1} \times 4C}$. It is converted to $\hat{C} \in \mathbb{R}^{H_l \times W_l \times C}$ to call back the resolution in the following way:

$$\hat{C}\left[2x+i, 2y+j, c\right] = \tilde{C}\left[x, y, (2i+j)c\right] \quad \forall i, j \in \{0, 1\} \tag{20}$$

where c denotes the class label and C denotes the number of classes. With the semantic context and feature sparsity hybrid encoding module, we can infer target classes more accurately, especially for targets that are texture-less due to speckle noise and for targets with heavy occlusion in complex dense scenes.

For regression task, it is more fine-grained and relies on boundary information and texture details. As mentioned above, existing approaches usually employ a single-scale feature map (e.g., P_l) to locate targets. However, feature maps at lower pyramid levels contain more edge, detail texture, and contour information. In addition, higher level feature maps can provide a global view for target localization. In accordance with this, we propose a detail and texture encoding module for precise localization. As depicted in Fig. 4(b), this module receives the feature maps from three neighboring scales as inputs. The fusion of P_{l+1} and P_{l-1} utilizes a simple U-Net [51] structure, which brings efficiency gains. P_l is first $2\times$ upsampled then aggregated with P_{l-1} to produce K_{l-1}. This design effectively preserves the texture and detail information in P_{l-1}. Finally, P_{l+1} is $2\times$ upsampled then aggregated with P_l and K_{l-1} to generate regression-oriented features D_l^{loc}. Which can be expressed as

$$D_l^{\text{loc}} = P_l + \text{u}(P_{l+1}) + \text{DConv}(\text{u}(P_l) + P_{l-1}) \tag{21}$$

where $\text{u}(\cdot)$ denotes the upsampling operation. D_l^{loc} is then projected by depthwise seperable convolutions and decoded by 1×1 convolutions to obtain the final predicted bounding boxes. It should be mentioned that the calculation of D_3^{loc} employs C_2, P_3, and P_4, since calculating P_2 through the neck leads to a huge computational burden.

IV. EXPERIMENT

This section validates the effectiveness of the proposed components and the superiority of DAFDet on four widely adopted SAR target detection datasets. First, implementation details are presented, including dataset information, evaluation metrics, and experimental setups. Next, ablation experiments are introduced to confirm the validity of the methodology. Finally, numerous comparisons are conducted with other methods.

A. Implementation Details

1) Datasets: To comprehensively evaluate the performance of DAFDet, we conduct extensive experiments on four publicly available SAR image target detection datasets, covering different sensor data, resolutions, polarizations, scenes, and targets with different scales and categories. Table I presents a quantitative comparison of these four datasets, indicating the purpose of the selection. Next, their individual properties will be described separately.

SSDD [52] is the first publicly available SAR image ship target detection dataset, which contains images of ship targets in different scenarios and sea states. The resolution is 1∼15 m. Furthermore, the SSDD includes a total of 1160 images and 2456 ship targets, most of which are small. It is the smallest, but most frequently employed research dataset.

MSAR-1.0 [53] is a multicategory SAR image experimental dataset, which covers a wide range of scenarios such as airports, harbors, near-shore, islands, offshore, and urban areas. It includes a total of 28 449 images for four categories of targets, of which 39 858 ships, 6368 airplanes, 12 319 oil tanks, and 1851 bridges. Some of the images have a size of 2048 × 2048 pixels, which facilitates the study of target detection in large-scale SAR images.

SAR-Ship-Dataset [54] contains a total of 43 819 multimode, multisource SAR images, of which there are 59 535 ships in total. Moreover, its resolution is 3∼25 m. Meanwhile, it contains a large number of multiscale ship targets and complex backgrounds, allowing for a more rigorous evaluation and comparison of the algorithms.

LS-SSDD-v1.0 [55] contains a total of 15 SAR images with 24 000 × 16 000 pixels from the Sentinel-1 satellite. The 15 large-scale SAR images are sliced into 9000 small-scale image slices to facilitate training and testing. In addition, it is a dataset geared toward large-scale ocean observation and close to practical engineering applications.

2) Evaluation Metrics: This article employs evaluation metrics from the COCO dataset [56] to be able to quantitatively analyze the performance of DAFDet and to facilitate fair

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comparisons with other methods. Where precision indicates the proportion of correctly predicted targets out of all targets predicted by the model. Recall represents the proportion of targets correctly predicted by the model among all GT targets. They are defined as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{22}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{23}$$

where false negative (FN), false positive (FP), and true positive (TP) are defined as in literatures [11], [15], [25]. Since it is difficult to judge the performance of the model with precision and recall alone, $F1$ is introduced to combine the measures. It is defined below

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{24}$$

Based on the recall and precision of a class, the average precision (AP) is defined in the following way:

$$\text{AP} = \int_0^1 p(r)\, dr \tag{25}$$

where r and p denote recall and precision, respectively. That is, the AP of this category is the area under its precision–recall curve. Accordingly, we can obtain AP^{50} and AP^{75}. The $\text{AP}^{50:95}$ is obtained by averaging the 10 values between AP^{50} and AP^{95} in a step of IoU = 0.05. Generally, $\text{AP}^{50:95}$ is denoted as AP. Thus, in this article, when denoting the overall performance of a model, AP refers to $\text{AP}^{50:95}$. Otherwise, it refers to the AP of a category. AP^{50} and AP^{75} are two significant metrics for evaluating the different capabilities of a model in target detection. AP^{50} is an effective indicator for evaluating the classification capability of the network, while AP^{75} can reflect the bounding box regression ability of the model. The AP index is the main consideration in the comparisons below, while referencing the AP^{50} and AP^{75} indexes.

3) Experiment Settings: For the experiments conducted on LS-SSDD-v1.0 and SSDD, they are both based on Intel[1] Core[2] i7-10700 CPUs and RTX 3080Ti GPUs. For the experiments performed on SAR-Ship-Dataset and MSAR-1.0, they are both based on a CentOS 7.6 operating system with AMD EPYC 7H12 CPUs and NVIDIA A100-SXM4-80GB GPUs. Except for LS-SSDD-v1.0 which follows the official data division, the other three datasets are all randomly divided into training, validation, and test sets in the ratio of 7:2:1. For the one-stage detector baseline, we employ some of the YOLOv7 [26] series models. For the two-stage detector baseline, we adopt the Faster R-CNN [57] model in the CBNet [34] framework with ResNet [50] as the backbone. The dynamic detectors and the corresponding baselines are trained with the same set of hyper-parameters. We train the Decision Maker for 2 epochs on a single GPU with batch = 1 using the AdamW [58] optimizer. "Dy-YOLOv7/10" indicates the dynamic YOLOv7 model with 10% of the SAR images classified as "hard" and the rest classified as "easy." The target confidence threshold and the IoU threshold for NMS are set to 0.001 and 0.65, respectively. In addition, FPS is measured with FP16-precision and batch = 1 on a single 3080Ti or A100 GPU, excluding postprocessing time.

[1]Registered trademark.
[2]Trademarked.

B. Ablation Studies

Extensive ablation experiments are carried out on SSDD and MSAR-1.0 datasets with different distributions and difficulties to fully validate the effectiveness of the proposed methods. The overall results of the ablation experiments are summarized in Table II. It can be seen that all the proposed methods contribute on both datasets with respect to the baselines. In the following, their different contributions will be analyzed in detail and the collaboration of the proposed components will be further discussed.

1) Effect of Dynamic Architecture: As indicated in Table II, the accuracy of SAR target detection achieves a significant improvement when YOLOv7 and YOLOv7-X are converted to dynamic models. For instance, on the SSDD dataset, Dy-YOLOv7/10 improves AP from 0.561 to 0.605. Dy-YOLOv7/50 improves it to 0.624. While the Dy-YOLOv7/100 directly improves it to 0.647, with an improvement of 8.6%. On the MSAR-1.0 dataset, Dy-YOLOv7-X/10 boosts the AP from 0.577 to 0.638. Dy-YOLOv7-X/50 boosts AP to 0.645. While the Dy-YOLOv7-X/100 directly boosts AP to 0.654, with a 7.7% improvement. Simultaneously, the inference speed is inevitably reduced, but still capable of real-time performance. For example, Dy-YOLOv7/10 achieves an inference speed of 85.47 FPS. Dy-YOLOv7/100 gives an inference speed of 60.61 FPS. The Dy-YOLOv7-X/50 also hits 48.08 FPS. This fully demonstrates the effectiveness of the developed dynamic architecture. Dynamic processing based on SAR image difficulty score is the key to its excellent effect. Fig. 5(a) visualizes SAR images with different difficulty scores. It can be clearly seen that different SAR images do vary in difficulty due to differences in target scale, target density, and background complexity. "Easy" SAR images contain fewer targets and have a clean background. Therefore, the multiscale features extracted by B1 are sufficient to be transmitted to the later layers to enable accurate localization and classification of the target. The "hard" SAR images contain obviously more complex scenes, with large target scale variations and severe occlusions, more small targets, and complex backgrounds. In this case, it is already difficult to attain satisfactory target localization and classification effects with only the multiscale features extracted by B2. While feeding the SAR image into B2, the feature representation capability is enriched to a certain degree by composing the features from all the higher level stages in B1 and fusing them into the lower level stages in B2, which ultimately improves the accuracy of target localization and classification. Determining the thresholds for classifying SAR images as "easy" and "hard" is the basis for the dynamic processing described above. Fig. 5(b) and (c) exhibits the change curves of difficulty scores on the validation sets for both SSDD and MSAR-1.0. Which demonstrates the effectiveness of the threshold determination strategy for the dynamic architecture by comparing the consistency of thresholds between validation

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TABLE II
ABLATION EXPERIMENTS. ALL THE RESULTS ARE OBTAINED WITH BATCH = 1 AND FP16-PRECISION ON A SINGLE RTX 3080TI OR A100

Datasets	Methods	Size	Precision	Recall	AP (%)	AP^{50} (%)	AP^{75} (%)	FPS	GFLOPs	Parameters
SSDD	YOLOv7 baseline	640	0.866	0.825	0.561	0.894	0.619	144.93	103.2	36.48M
	Dy-YOLOv7/10	640	**0.961**	0.843	0.605	0.949	0.702	85.47	**111.3**	78.34M
	+ AFEFPN	640	0.943	**0.937**	0.659	0.974	0.787	**95.24**	124.5	**61.44M**
	+ TODH	640	0.947	0.935	**0.663**	**0.975**	**0.801**	63.69	140.6	91.18M
	Dy-YOLOv7/50	640	**0.959**	0.873	0.624	0.951	0.721	68.03	**142.1**	78.34M
	+ AFEFPN	640	0.940	**0.937**	0.665	0.973	0.799	**76.92**	155.3	**61.44M**
	+ TODH	640	0.952	0.934	**0.672**	**0.976**	**0.823**	57.80	172.5	91.18M
	Dy-YOLOv7/90	640	0.932	0.877	0.632	0.951	0.745	64.94	**172.2**	78.34M
	+ AFEFPN	640	0.961	0.929	0.669	**0.976**	0.800	**66.67**	185.4	**61.44M**
	+ TODH	640	**0.964**	**0.935**	**0.676**	0.973	**0.827**	50.76	202.3	91.18M
	Dy-YOLOv7/100	640	0.944	0.877	0.647	0.956	0.760	60.61	**182.0**	78.34M
	+ AFEFPN	640	0.966	0.944	0.674	0.977	0.798	**65.79**	195.2	**61.44M**
	+ TODH	640	**0.969**	**0.947**	**0.683**	**0.981**	**0.824**	50.25	208.7	91.18M
MSAR-1.0	YOLOv7-X baseline	640	0.846	0.865	0.577	0.898	0.621	119.05	188.0	70.80M
	Dy-YOLOv7-X/10	640	0.889	0.889	0.638	0.925	0.701	69.93	198.9	150.00M
	+ AFEFPN	640	0.928	0.942	0.788	0.960	0.857	**78.13**	**160.7**	**94.77M**
	+ TODH	640	**0.933**	**0.946**	**0.799**	**0.964**	**0.861**	67.54	169.9	120.15M
	Dy-YOLOv7-X/50	640	0.881	0.897	0.645	0.925	0.708	48.08	246.9	150.00M
	+ AFEFPN	640	0.936	0.946	0.800	0.963	0.866	**67.57**	**208.7**	**94.77M**
	+ TODH	640	**0.942**	**0.947**	**0.809**	**0.967**	**0.868**	56.21	222.5	120.15M
	Dy-YOLOv7-X/90	640	0.861	0.918	0.653	0.930	0.725	40.49	294.9	150.00M
	+ AFEFPN	640	0.948	0.936	0.807	0.965	0.874	**57.80**	**256.7**	**94.77M**
	+ TODH	640	**0.951**	**0.939**	**0.818**	**0.968**	**0.876**	45.37	271.3	120.15M
	Dy-YOLOv7-X/100	640	0.863	0.919	0.654	0.929	0.730	37.74	306.8	150.00M
	+ AFEFPN	640	0.950	0.940	0.810	0.967	0.878	**56.82**	**268.6**	**94.77M**
	+ TODH	640	**0.953**	**0.944**	**0.822**	**0.972**	**0.880**	42.96	276.7	120.15M

TABLE III
COMPATIBILITY FOR TWO-STAGE DETECTORS. THE RESULTS ARE OBTAINED WITH BATCH = 1 AND FP16-PRECISION ON A SINGLE A100

Methods	AP (%)	FPS	GFLOPs
Faster R-CNN(ResNet50)	0.483	28.61	210.5
Faster R-CNN(ResNet101)	0.512	22.13	285.7
Dynamic Faster R-CNN(ResNet50)/50	0.515	24.18	248.1
Dynamic Faster R-CNN(ResNet50)/90	0.523	20.87	277.6

and test sets on SSDD and MSAR-1.0. As an example, Dy-YOLOv7 on the SSDD dataset in Fig. 5(b) has a threshold of 0.4647 for the 10% quantile on the test set and 0.4658 for the 10% quantile on the validation set. Its threshold for the 90% quantile on the test set is 0.4856, while the threshold for the 90% quantile on the validation set is 0.4822. Different degrees of inconsistency emerge between the validation set thresholds and the test set thresholds under the above two quantiles. In contrast, Dy-YOLOv7-X on MSAR-1.0 in Fig. 5(c) fully demonstrates the consistency of thresholds between the validation and test sets. The main reason for this is the different dataset sizes and distributions. SSDD is very small and unevenly distributed, mainly concentrated on small and medium-sized targets. MSAR-1.0 is large and evenly distributed, thus the thresholds on the validation set can be well generalized to the test set.

In addition, the dynamic architecture is compatible with two-stage detectors. With the classical two-stage detector Faster R-CNN as the baseline, relevant experiments are conducted on MSAR-1.0, as presented by Table III. Experimental results reveal that with the utilization of the dynamic architecture, the AP is improved by 3.2% and 4%, respectively. Compared to the ResNet101 backbone-based Faster R-CNN, Dynamic Faster R-CNN(ResNet50)/90 reduces the computational complexity by 8.1 GFLOPs and improves the accuracy by 1.1%, fully demonstrating the generality of the dynamic architecture.

2) Effect of AFEFPN: The proposed AFEFPN attains significant improvements over all baselines, as displayed by Table II. For example, with the application of AFEFPN to Dy-YOLOv7/10 on SSDD, the AP is improved from 60.5% to 65.9%, with a 5.4% improvement. The detection efficiency increases from 85.47 to 95.24 FPS, and the number of parameters decreases from 78.34M to 61.44M. After applying AFEFPN to Dy-YOLOv7-X/90 on MSAR-1.0, AP is improved from 65.3% to 80.7%, resulting in a 15.4% improvement. Meanwhile, the detection efficiency increases from 40.49 to 57.8 FPS, and the time complexity and space complexity decrease from 294.9G and 150M to 256.7G and 94.77M, respectively. Following the application of AFEFPN to the Dy-YOLOv7-X/100, AP is improved from 65.4% to 81%, with an improvement of 15.6%. The detection efficiency increases from 37.74 to 56.82 FPS, and the time complexity decreases from 306.8G to 268.6G. All these sufficiently demonstrate the validity of the proposed AFEFPN. Additional experiments are carried out to further illustrate its effect. As presented in Table IV, the proposed AFEFPN exhibits overwhelming advantages over other widely employed FPNs. Rep-PAN [39] improves the AP of the baseline (PAFPN [37]) by only 0.1%. In comparison, BiFPN [38] offers more powerful performance, which improves AP by 2.2% with the lowest time complexity (95.8G) and satisfactory space complexity (33.13M). The AFPN [41] is a highly competitive endeavor that improves AP by 10% from baseline. However, it suffers from the highest

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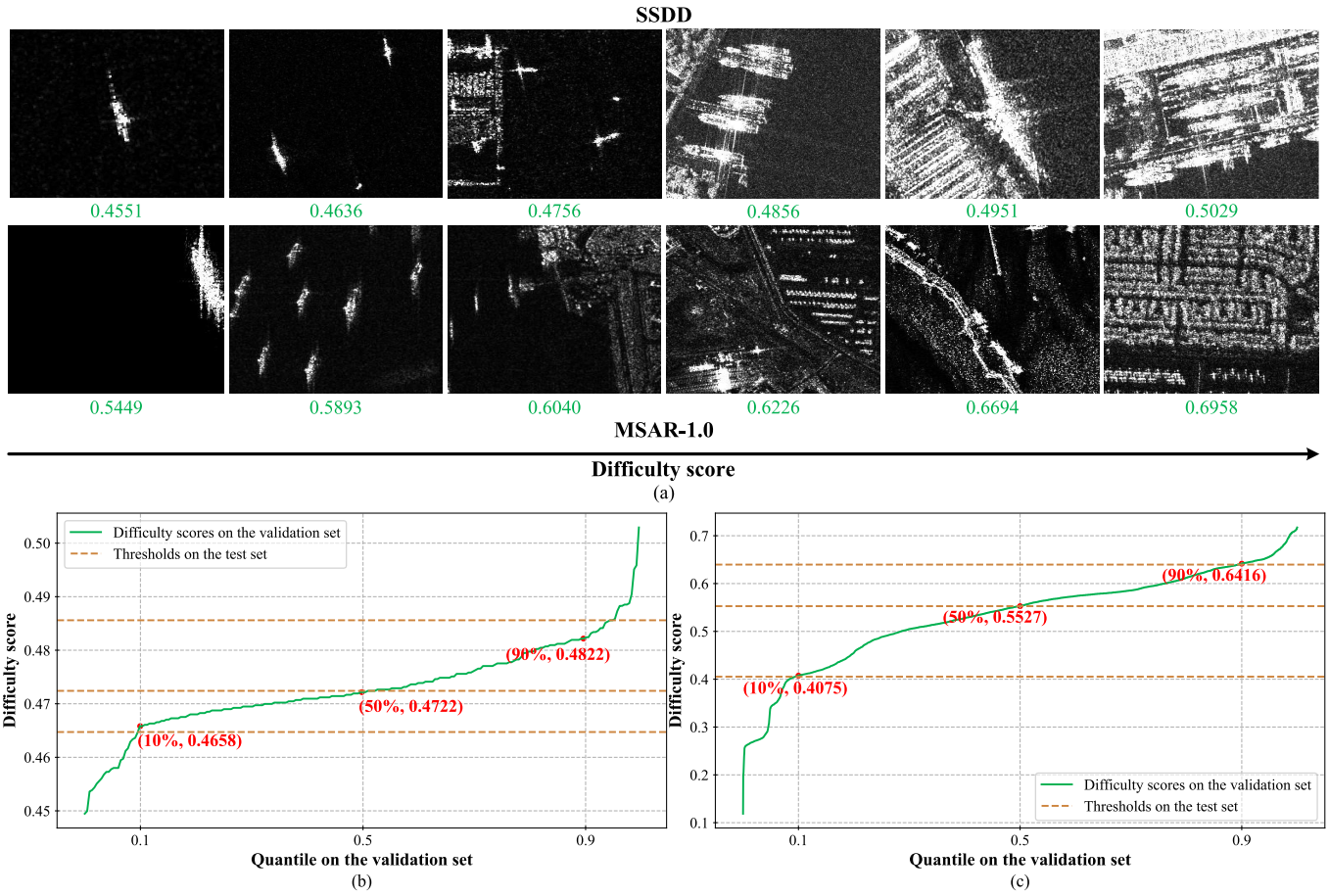


Fig. 5. Visualizations of SAR images with different difficulty scores and change curves for difficulty scores. (a) Visualizations of "easy" and "hard" SAR images. The horizontal direction indicates the difficulty scores predicted by the decision makers in Dy-YOLOv7 and Dy-YOLOv7-X on the corresponding datasets, which are larger the further to the right. (b) Low-to-high change curve of difficulty scores on the validation set of SSDD. (c) Low-to-high change curve of difficulty scores on the validation set of MSAR-1.0.

TABLE IV

CHALLENGE TO OTHER FEATURE PYRAMIDS. THE RESULTS ARE OBTAINED WITH BATCH = 1 AND FP16-PRECISION ON A SINGLE 3080TI OR A100

Datasets	Methods	Size	Precision	Recall	AP (%)	AP^{50} (%)	AP^{75} (%)	FPS	GFLOPs	Parameters
SSDD	YOLOv7(PAFPN [37])	640	0.866	0.825	0.561	0.894	0.619	144.93	103.2	36.48M
	+ AFPN [41]	640	0.954	0.937	0.661	0.968	0.793	107.53	195.0	61.00M
	+ Rep-PAN [39]	640	0.866	0.817	0.562	0.889	0.669	104.17	138.5	50.44M
	+ BiFPN [38]	640	0.894	0.821	0.583	0.920	0.685	142.86	**95.8**	33.13M
	+ AFEFPN(ours)	640	**0.973**	**0.944**	**0.697**	**0.975**	**0.863**	**147.06**	109.8	**28.03M**
	YOLOv7-X(PAFPN [37])	640	0.920	0.899	0.628	0.958	0.747	112.36	188.0	70.78M
	+ AFPN [41]	640	0.941	0.907	0.653	0.967	0.809	84.75	302.0	94.67M
	+ AFEFPN(ours)	640	**0.988**	**0.959**	**0.694**	**0.985**	**0.837**	**121.95**	**168.8**	**43.16M**
MSAR-1.0	YOLOv7(PAFPN [37])	640	0.915	0.927	0.703	0.952	0.776	**131.58**	**103.2**	36.50M
	+ AFPN [41]	640	0.922	**0.937**	**0.739**	**0.954**	**0.814**	101.01	195.1	61.02M
	+ AFEFPN(ours)	640	**0.925**	0.921	0.738	0.952	0.813	129.87	109.9	**28.05M**
	YOLOv7-X(PAFPN [37])	640	0.846	0.865	0.577	0.898	0.621	119.05	188.0	70.80M
	+ AFPN [41]	640	**0.943**	**0.927**	**0.782**	**0.961**	**0.844**	81.97	302.0	94.69M
	+ AFEFPN(ours)	640	0.923	0.926	0.757	0.953	0.824	**123.46**	**168.9**	**43.18M**

time complexity (195G) and space complexity (61M), and a 37.4 FPS decrease in inference speed. In addition to an increase of only 6.6G in time complexity, our AFEFPN delivers the best results in terms of accuracy, inference speed, and space complexity, with 69.7% AP, 147.06 FPS, and 28.03M, respectively. It is worth mentioning that the inference speed increases by 2.13 FPS and the space complexity decreases by 8.45M with the AFEFPN. The same overwhelming results are obtained on the YOLOv7-X with the PAFPN [37] as the neck. On the MSAR-1.0 dataset, our AFEFPN attains comparable accuracy to the AFPN [41], but performs better with respect to detection efficiency, time complexity, and space complexity. Taking YOLOv7-X as an example, the accuracy with AFEFPN is only 2.5% lower than that with AFPN [41], but 133.1G and 51.51M lower in terms of time complexity and space complexity, respectively, and 41.49 FPS higher

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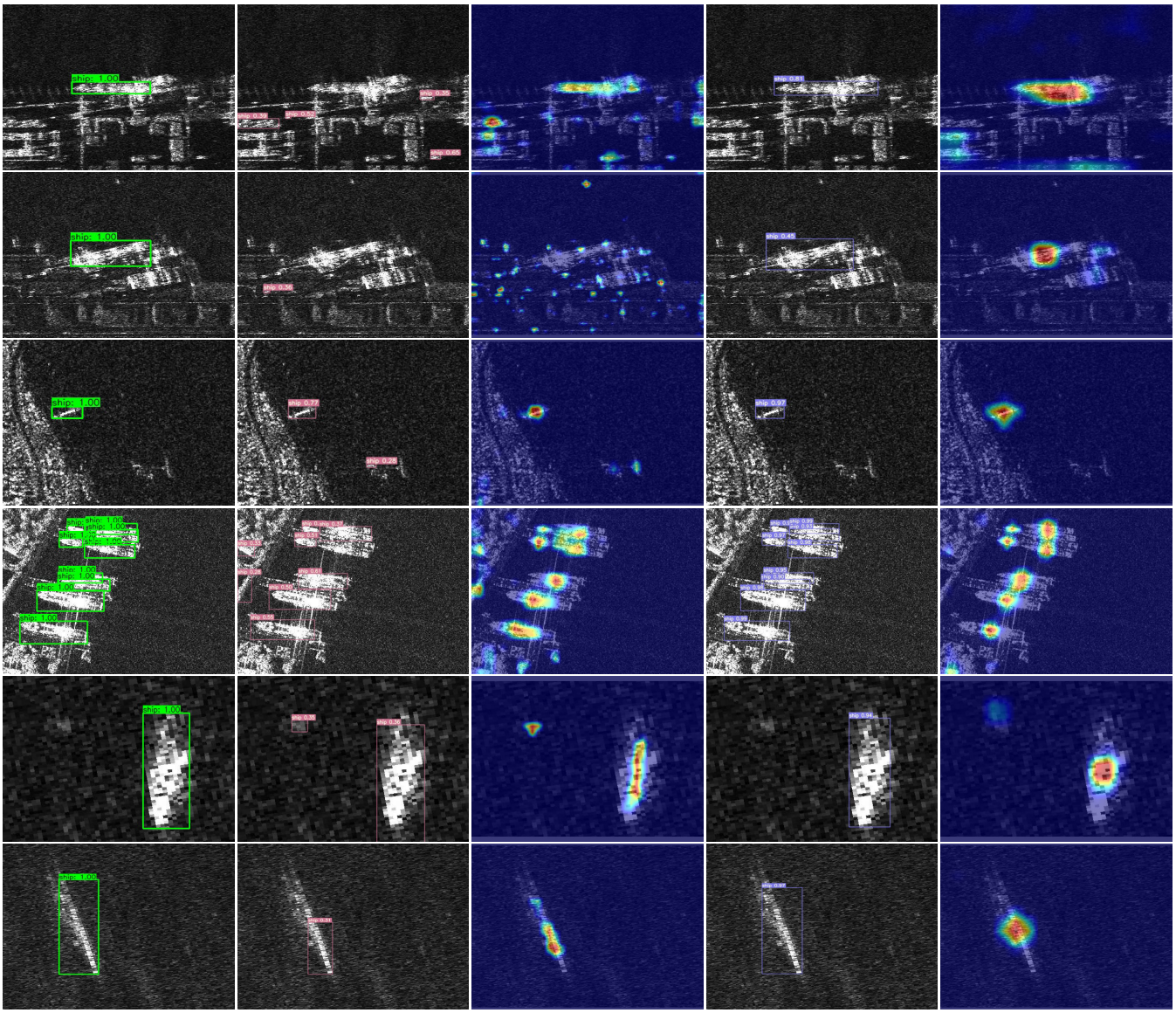
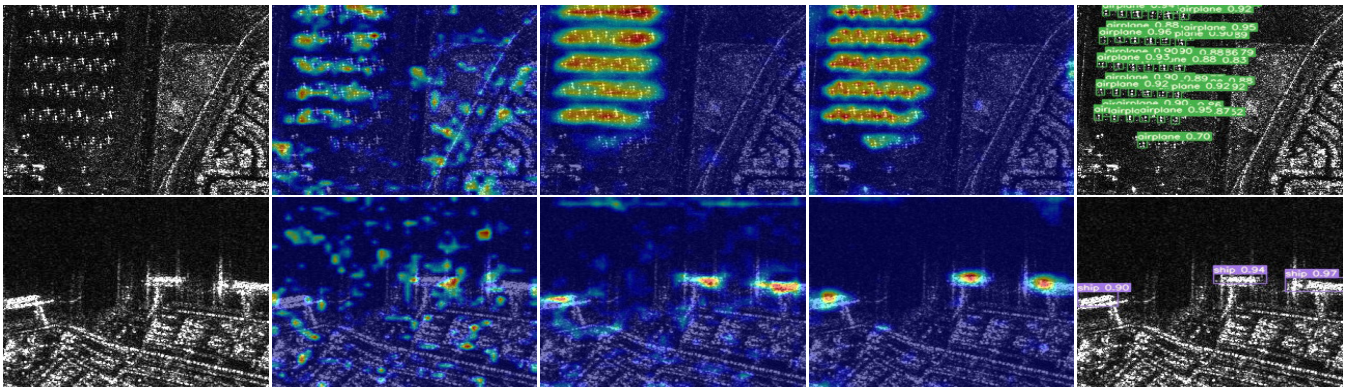


Fig. 6. Comparison of detection effects and attention heatmaps without and with the AFEFPN module. (a) GT. (b) Detection results without the AFEFPN module. (c) Attention heatmaps without the AFEFPN module. (d) Detection results with the AFEFPN module. (e) Attention heatmaps with the AFEFPN module.

in terms of inference speed. All of these strongly confirm the power of the AFEFPN. Asymptotic architecture, adaptive spatial fusion, channel enhancement, and compression of the number of channels are the keys to its effectiveness. The fusion scheme of AFEFPN allows for direct fusion of information from different levels of features, avoiding degradation or loss of information, while effectively reducing the large semantic gap between nonadjacent layers. Ultimately, the asymptotic architecture allows significant multiscale features to contain both abundant raw semantic and detailed information.

The adaptive spatial fusion operation enhances the importance of critical level features by adaptively assigning level-dependent spatial weights at each spatial location, effectively mitigating the problem of multiobject information conflict, and ultimately retaining more useful information for fusion. On top of the adaptive spatial feature fusion, the channel enhancement operation effectively suppresses the background noise by disentangling it from the target information on the feature channel, thus allowing the model to focus more on the target. The visualizations of the ablation experiments on the AFEFPN module in Fig. 6 sufficiently demonstrate the capability of the network to efficiently extract target features and suppress background noise. Fig. 6(a) indicates the GT. Fig. 6(b) and (c) represents the detection results and the corresponding attention heatmaps without the AFEFPN module, respectively. Fig. 6(d) and (e) denotes the detection results and the corresponding attention heatmaps with the utilization of the AFEFPN module, respectively. From the comparisons of the detection results for the first, second, and fourth rows, it can be seen that the utilization of the AFEFPN module can effectively reduce the missing alarms and thus increase the recall. This is fully consistent

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(a) (b) (c) (d) (e)

Fig. 7. Visualizations of the feature extraction effects. (a) Represents the raw SAR images, (b)–(d) denote the heatmaps generated by the input feature layer C_3, the intermediate feature layer, and the output feature layer P_3 of the AFEFPN, respectively, and (e) indicates detection results.

TABLE V

COMPARISONS WITH DIFFERENT DETECTION HEADS. ALL THE RESULTS ARE OBTAINED WITH BATCH = 1 AND FP16-PRECISION ON A SINGLE RTX 3080TI OR A100

Datasets	Methods	Size	Precision	Recall	AP (%)	AP^{50} (%)	AP^{75} (%)	FPS	GFLOPs	Parameters
SSDD	YOLOv7(Coupled Head [44])	640	0.866	0.825	0.561	0.894	0.619	**144.93**	**103.2**	**36.48M**
	+ YOLOX Decoupled Head [13]	640	0.898	0.951	0.653	0.957	0.791	104.17	144.2	44.00M
	+ Efficient Decoupled Head [39]	640	**0.984**	0.933	0.659	0.951	0.809	112.36	151.0	62.64M
	+ TSCODE [45]	640	0.941	0.955	**0.687**	0.977	**0.843**	89.29	136.1	71.81M
	+ TODH(ours)	640	0.955	**0.959**	0.666	**0.986**	0.808	90.09	109.7	49.15M
MSAR-1.0	YOLOv7(Coupled Head [44])	640	0.915	0.927	0.703	0.952	0.776	**131.58**	**103.2**	**36.50M**
	+ YOLOX Decoupled Head [13]	640	0.917	0.949	0.738	0.952	0.806	97.09	144.3	44.01M
	+ Efficient Decoupled Head [39]	640	0.904	0.956	0.740	0.954	0.819	101.01	151.0	62.66M
	+ TSCODE [45]	640	**0.951**	0.947	**0.817**	**0.969**	**0.878**	74.63	136.1	71.84M
	+ TODH(ours)	640	0.942	**0.957**	0.788	**0.969**	0.859	80.65	109.7	49.19M

with the quantitative results on recall in Tables II and IV. Comparisons of the detection results from the first five rows show that false alarms caused by background noise are perfectly eliminated, resulting in improved precisions. This also coincides with the quantitative results on precision in Tables II and IV. The comparisons of the detection results in rows 5 and 6 indicate that the localization effect of the target is significantly better with the AFEFPN module. In addition, the comparisons of the heatmaps in Fig. 6 reveal that with the AFEFPN module, the model is more focused on the target, is difficult to interfere by the background noise, and has a higher confidence for the prediction result. All these stem from effective target information extraction derived from asymptotic architecture and adaptive spatial feature fusion and background noise suppression derived from channel enhancement. This is further illustrated by the visualizations of the heatmaps for different feature layers in Fig. 7. As shown in the first row, for the raw SAR image presented in Fig. 7(a) and (b) reveals that the feature map C_3 extracted by the backbone has not yet contained effective target information and contains substantial background interference information. When C_3 is fed into the AFEFPN module and undergoes twice adaptive spatial feature fusion and channel enhancement operations, heatmaps of different feature maps are generated sequentially as depicted in Fig. 7(c) and (d). They show that effective target information is extracted and that both semantic and detailed information is contained in the extracted features. Finally, the densely packed targets are accurately detected, as shown in Fig. 7(e). For the background noise interference, the AFEFPN module gradually disentangles it from the target by channel enhancement operation, and ultimately realizes the background noise suppression, as indicated in the second row of Fig. 7. While the compression of the number of channels can reduce both the number of parameters and the amount of computation, thus leading to an increase in the inference speed.

TABLE VI

CHALLENGING UNIVERSAL DETECTION ALGORITHMS ON SSDD

Methods	Backbone	AP (%)	AP^{50} (%)	AP^{75} (%)
Anchor-based(one-stage)				
YOLOv3 [43]	DarkNet-53	0.624	0.962	0.721
RetinaNet [59]	ResNet-50	0.664	0.961	0.783
SSD512 [60]	VGG-16	0.666	0.941	0.827
Anchor-based(two-stage)				
Faster R-CNN [57]	ResNet-50	**0.694**	0.953	0.857
Cascade R-CNN [61]	ResNet-50	0.677	0.918	**0.862**
Anchor-free(one-stage)				
CenterNet [14]	ResNet-18	0.622	0.942	0.728
CornerNet [62]	Hourglass-104	0.654	0.921	0.775
CenterNet [14]	DLA-34	-	0.946	-
YOLOX-S [13]	CSPDarknet-53	0.673	0.971	0.784
FCOS [63]	ResNet-50	0.686	0.970	0.832
DAFDet-YOLOv7/10	Dual-Backbone	0.663	**0.975**	0.801
DAFDet-YOLOv7/50	Dual-Backbone	0.672	**0.976**	0.823
DAFDet-YOLOv7/90	Dual-Backbone	0.676	**0.973**	0.827
DAFDet-YOLOv7/100	Dual-Backbone	0.683	**0.981**	0.824

3) Effect of TODH: The TODH is designed to decouple the feature encoding at the source thereby learning different

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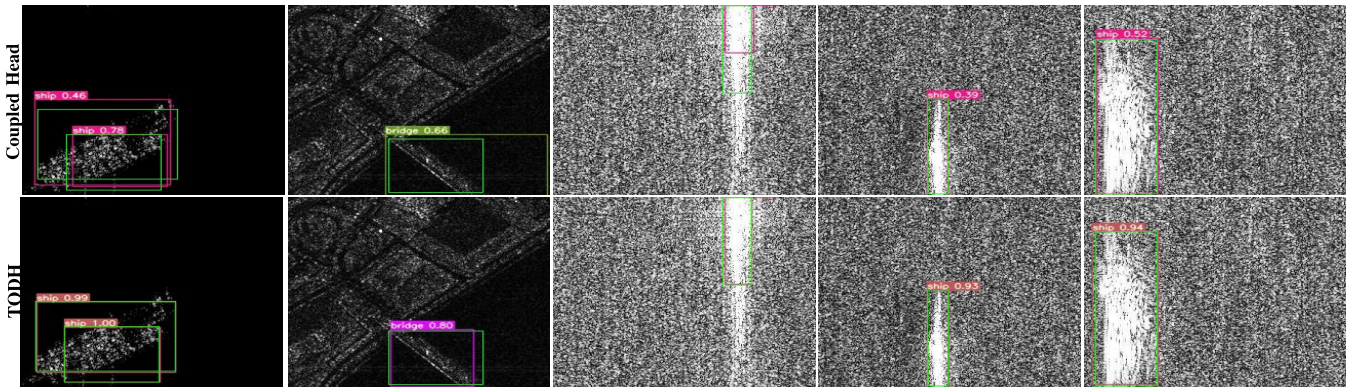


Fig. 8. Comparisons of the bounding box regression and the classification confidence with and without the TODH module. The green boxes indicate GT bounding boxes.

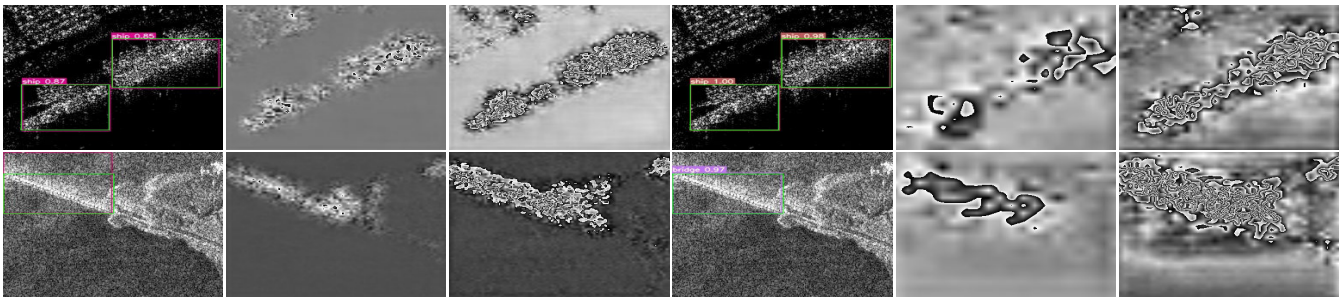


Fig. 9. Visualizations of feature maps extracted by different branches. (a) Detection results of YOLOX Decoupled Head [13]. (d) Detection results of our TODH. (b) and (c) and (e) and (f) Feature maps extracted by the classification and regression branches of YOLOX Decoupled Head [13] and TODH, respectively. The green boxes indicate GT bounding boxes.

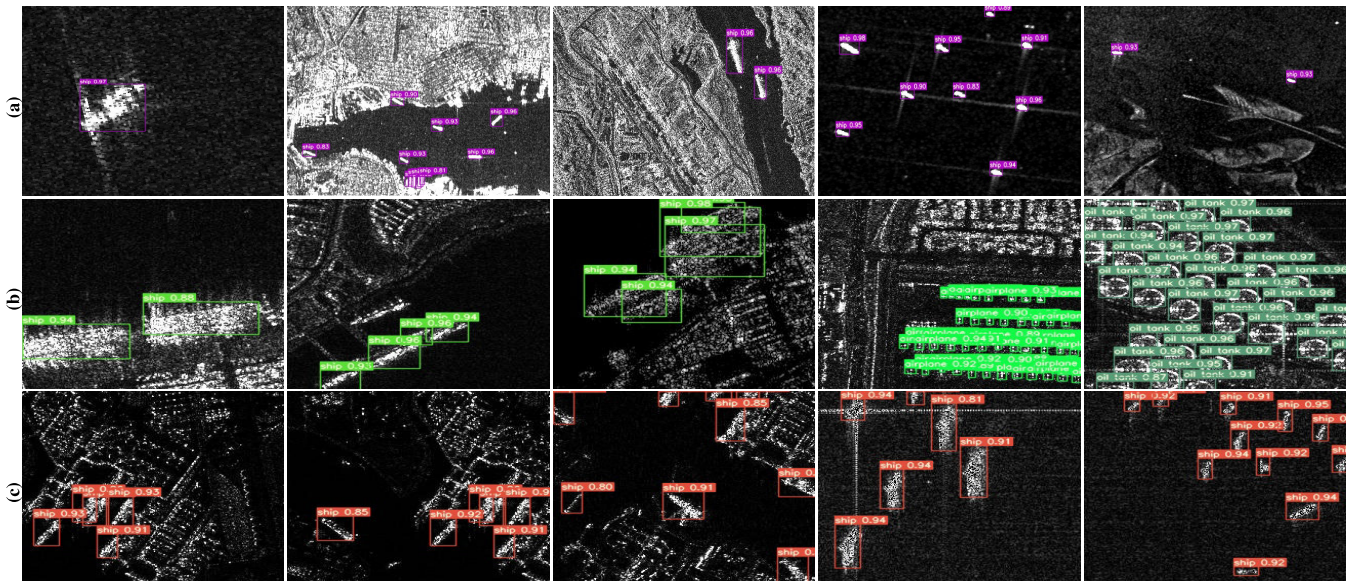


Fig. 10. Qualitative identification results of the proposed DAFDet on (a) SSDD, (b) MSAR-1.0, and (c) SAR-Ship-Dataset datasets, respectively. Boxes with different colors indicate targets of different categories.

feature contexts for classification and regression tasks. As displayed in Table II, upon replacing the Coupled Head [44] with the proposed TODH, the detection accuracy achieves different degrees of improvements on both datasets. For example, with the TODH, the AP on the SSDD is improved from 67.4% to 68.3%, and the AP on the MSAR-1.0 is improved from 81% to 82.2%. More extensive experiments are conducted to further illustrate its effect. As presented in Table V, the proposed TODH improves the detection accuracy by 10.5% and 8.5% on SSDD and MSAR-1.0 datasets, respectively. In addition, it exhibits the best classification capability and recall, excellent bounding box regression capability, and precision. Simultaneously, it provides considerably low time complexity and space complexity as well as efficient inference speed. In contrast, though the Coupled Head [44] presents the lowest time and space complexity as well as the most

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efficient inference speed, it suffers from the poorest detection accuracy. YOLOX Decoupled Head [13] obtains some gains in detection accuracy, but the increase in time complexity is also highly noticeable. Efficient Decoupled Head [39] offers the highest time complexity and is also quite high in space complexity. TSCODE [45] is an extremely competitive effort with the highest detection accuracy. However, it suffers from the highest space complexity, the most inefficient inference speed, and an increase in time complexity approaching 33G. Overall, the proposed TODH achieves the best performance. Fig. 8 qualitatively demonstrates the contributions of the TODH. As indicated in the first and second columns, both classification confidence scores and bounding box regression accuracies are substantially improved with the TODH. The third column further illustrates this point. Even with similar bounding box regression accuracies, the classification confidence scores are significantly improved with the TODH, as shown in the fourth and fifth columns. The fundamental reason is that regression and classification tasks have different preferences for feature context. Classification usually favors more semantic features, while regression expects more boundary-aware features such as detail and texture. The Coupled Head [44] employs the same convolution to encode both semantic and boundary-aware information. YOLOX Decoupled Head [13] and Efficient Decoupled Head [39] utilize disentangled heads to learn different feature contexts for different tasks. However, the learned feature context is still based on the same input feature at the source. They all introduce a degree of conflict between the classification and regression tasks. While the proposed TODH first decouples the feature encoding at the source, thus providing task-oriented feature contexts. Specifically, the semantic context and feature sparsity hybrid coding module is employed to learn spatially coarser but semantically richer feature maps. The detail and texture encoding module is utilized to construct feature maps containing detailed texture and boundary information. On the basis of the decoupled feature context, the classification branch can construct richer semantic information to accurately infer the category of the target, while the regression branch can model more detail and texture information to accurately regress the target boundary. This is amply borne out by Fig. 9. The developed TODH constructs richer semantic features and more detail and texture features on the classification and regression branches, respectively, thus enabling the model to obtain higher classification confidence and more accurate bounding box regression.

4) Collaborations of Different Components: Upon analyzing the effects of individual components, we further discuss the collaborations of different components. Aiming at the general difficulties of current SAR target detection methods in balancing speed and accuracy and in extracting effective target information from complex scenes, this article proposes dynamic architecture, AFEFPN and TODH, to address these challenges. The proposed three components are independent, and orthogonal and do not affect each other. Table II demonstrates the collaboration of the dynamic architecture, AFEFPN and TODH on two baselines. The dynamic architecture first performs dynamic processing to determine the inference path based on the difficulty score of the input SAR image. Then, the AFEFPN performs asymptotic fusion enhancement of the multiscale features extracted from the backbone, thus outputting significant multiscale features that contain both abundant raw semantic and detail information and that are free of background noise. Finally, TODH decouples the above feature encodings at the source to generate classification predictions with rich semantic information and regression predictions with rich boundary-aware information. In this process, the dynamic architecture, AFEFPN and TODH effectively improve the accuracy of the model through dynamic processing, asymptotic fusion, and encoding decoupling, respectively, contributing to the overall effectiveness. As for other collaborations of components, please refer to Section B of the Supplementary Material. The performance impact and contribution of each component can be clearly seen, further illustrating their orthogonality.

TABLE VII
CHALLENGING UNIVERSAL DETECTION ALGORITHMS ON MSAR-1.0

Methods	Backbone	AP (%)	AP^{50} (%)	AP^{75} (%)
DDOD [64]	ResNet-50	0.548	0.888	0.577
Deformable DETR [65]	ResNet-50	0.420	0.718	0.439
YOLOF [66]	ResNet-50	0.487	0.776	0.528
Sparse R-CNN [67]	ResNet-50	0.505	0.810	0.540
VarifocalNet [68]	ResNet-50	0.531	0.865	0.549
ATSS [69]	ResNet-50	0.536	0.867	0.548
FoveaBox [70]	ResNet-101	0.519	0.859	0.544
FSAF [71]	ResNeXt-101	0.448	0.824	0.436
Guided Anchoring [72]	ResNet-101	0.476	0.822	0.484
YOLOv7 [26]	ELAN	0.611	0.917	0.655
DAFDet-YOLOv7-X/10	Dual-Backbone	**0.799**	**0.964**	**0.861**
DAFDet-YOLOv7-X/50	Dual-Backbone	**0.809**	**0.967**	**0.868**
DAFDet-YOLOv7-X/90	Dual-Backbone	**0.818**	**0.968**	**0.876**
DAFDet-YOLOv7-X/100	Dual-Backbone	**0.822**	**0.972**	**0.880**

C. Comparison With State-of-the-Art Methods

In order to fully illustrate the superiority of the proposed DAFDet, we conducted extensive comparative experiments on three publicly available datasets, SSDD, MSAR-1.0, and SAR-Ship-Dataset. First, comparisons are carried out with DL-based methods that are currently being intensively investigated, and then with previously classical and conventional approaches.

1) Challenging DL-Based Algorithms:

a) Challenging universal algorithms: Universal algorithms generally developed upon optical data have also achieved favorable results in the field of target detection for SAR images [25]. To validate the advantages of DAFDet over these methods, we compared it with some most advanced universal detection algorithms on the multisource dataset SSDD, as presented in Table VI. It can be seen that DAFDet attains excellent accuracy. DAFDet-YOLOv7/100 achieves 68.3% AP, only 1.1% lower than Faster R-CNN [57]. Meanwhile, it offers the best AP^{50} of 98.1%. In addition, our methods outperform all comparison methods on AP^{50}. And, a single model, DAFDet-YOLOv7, can achieve detection effects with different accuracies only by adjusting the threshold. Fig. 10(a) visualizes the detection results of DAFDet-YOLOv7/100. It can be clearly seen that the proposed DAFDet has excellent detection accuracy, high classification, and localization accuracy, and is difficult to be interfered with by complex background noise.

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TABLE VIII
CHALLENGING UNIVERSAL DETECTION ALGORITHMS ON SAR-SHIP-DATASET

Methods	Backbone	Size	Precision	Recall	AP (%)	AP^{50} (%)	AP^{75} (%)	FPS (A100)	GFLOPs	Parameters
YOLOv7-E6 [26]	ELAN	1280	0.930	0.939	0.696	0.965	0.837	71.94	143.5	110.33M
YOLOv7-D6 [26]	ELAN	1280	0.929	0.943	0.698	0.966	0.840	60.24	197.3	152.83M
YOLOv7-E6E [26]	ELAN	1280	0.937	0.941	0.696	0.967	0.841	46.73	225.4	164.82M
YOLOv3 [43]	DarkNet53	640	-	-	0.614	0.948	0.722	-	77.54	61.52M
SSD-300 [60]	VGG16	640	-	-	0.610	0.937	0.711	-	137.09	23.75M
SSD-512 [60]	VGG16	640	-	-	0.636	0.946	0.756	-	137.07	24.39M
CenterNet [73]	ResNet18	640	-	-	0.559	0.937	0.608	-	**19.34**	**14.43M**
VarifocalNet [68]	ResNet50	640	-	-	0.624	0.948	0.720	-	75.56	32.48M
YOLOv7 [26]	ELAN	640	0.932	0.930	0.684	0.964	0.816	**125.00**	103.2	36.48M
YOLOv7-X [26]	ELAN	640	0.875	0.867	0.555	0.917	0.611	114.94	188.0	70.78M
DAFDet-YOLOv7/10	Dual-Backbone	640	0.934	0.943	0.687	0.966	0.828	76.92	140.6	91.18M
DAFDet-YOLOv7/50	Dual-Backbone	640	0.935	0.943	0.688	0.965	0.828	76.34	172.5	91.18M
DAFDet-YOLOv7/90	Dual-Backbone	640	0.939	0.937	0.692	0.965	0.831	67.57	202.3	91.18M
DAFDet-YOLOv7/100	Dual-Backbone	640	0.938	0.937	0.693	0.964	0.833	62.50	208.7	91.18M
DAFDet-YOLOv7-X/10	Dual-Backbone	640	0.937	0.937	0.694	**0.968**	0.835	76.34	169.9	120.13M
DAFDet-YOLOv7-X/50	Dual-Backbone	640	0.930	**0.945**	0.695	0.967	0.836	64.52	222.5	120.13M
DAFDet-YOLOv7-X/90	Dual-Backbone	640	0.942	0.933	0.698	0.966	0.842	56.82	271.3	120.13M
DAFDet-YOLOv7-X/100	Dual-Backbone	640	**0.943**	0.932	**0.700**	0.965	**0.844**	41.67	276.7	120.13M

TABLE IX
COMPARISON WITH STATE-OF-THE-ART SAR TARGET DETECTION METHODS ON SSDD

Methods	P	R	F1	mAP
AFSar [15]	0.941	**0.982**	**0.961**	0.977
CenterNet++ [16]	0.833	0.952	0.889	0.951
DCMSNN [17]	0.905	0.891	0.898	0.894
DSDet [18]	-	-	-	0.972
TWC-Net [19]	0.914	0.953	0.933	-
FBR-Net [20]	0.928	0.940	0.934	0.941
ISASDNet [21]	-	-	-	0.968
Key-Point Estimation + Channel Attention [22]	0.948	0.961	0.954	0.963
DiffDet4SAR [74]	0.935	0.969	0.952	0.966
MLSDNet [75]	-	-	0.937	0.972
HRLE-SARDet [76]	-	-	0.948	0.976
CMFT [77]	0.924	0.981	0.952	0.973
DAFDet-YOLOv7/100(ours)	**0.969**	0.947	0.958	**0.981**

TABLE X
CHALLENGING THE MOST ADVANCED SAR-SPECIFIC DETECTION ALGORITHMS ON SAR-SHIP-DATASET

Methods	mAP (%)	times (ms)
SSE-CenterNet [11]	94.1	12.5
DAPN [23]	87.9	71.2
Scale-transferrable Pyramid Network [24]	86.8	98.3
Key-Point Estimation + Channel Attention [22]	94.7	31.7
ADERLNet-CW [25]	95.8	9.9
DiffDet4SAR [74]	95.2	13.3
DAFDet-YOLOv7-X/10(ours)	**96.8**	**9.6**

TABLE XI
CHALLENGING TRADITIONAL ALGORITHMS ON SSDD

Methods	P	R	F1
OS-CFAR [6]	0.840	**0.988**	0.908
CA-CFAR [5]	0.861	0.982	0.917
CHA-CFAR [78]	0.872	0.985	0.925
ASL-CFAR [79]	0.881	0.983	0.929
DAFDet-YOLOv7/100(ours)	**0.969**	0.947	**0.958**

To further verify the advantages of the developed DAFDet for detecting multiclass targets, we carried out comparative experiments on the MSAR-1.0 dataset, as indicated in Table VII. It intuitively indicates that our methods achieve overwhelming superiority on both AP, AP^{50} and AP^{75}. Relative to the best-performing YOLOv7 [26], our method achieves a maximum improvement of 21.1%, 5.5%, and 22.5% for AP, AP^{50}, and AP^{75}, respectively. The quantitative statistics in Table VII are strongly supported by the qualitative detection results in Fig. 10(b). The first column shows that the model is hardly affected by degradation or loss of significant information associated with texture and detail caused by speckle noise. The second and third columns reveal that our method still provides strong classification and localization capabilities for densely packed targets with heavy occlusion. The fourth and fifth columns indicate that the model can accurately localize the target for scenarios with densely packed targets and with natural clutter, and does not produce false alarms due to natural clutter.

To validate the overall performance of DAFDet, more rigorous comparison experiments are conducted on the large-scale, multisource, and multimode dataset SAR-Ship-Dataset, as illustrated in Table VIII. Compared to models with an input size of 1280 × 1280, our methods achieve comparable accuracy with fewer parameters and faster inference. For example, DAFDet-YOLOv7-X/90 is 0.2% higher than YOLOv7-E6E [26] in terms of AP, 21.6% faster in terms of inference speed, and 27.1% fewer in terms of parameters. The DAFDet-YOLOv7-X/50 is 0.3% less accurate than the YOLOv7-D6 [26], but 7.1% faster in inference and 21.4% fewer in parameters. DAFDet-YOLOv7-X/10 is 0.2% lower than YOLOv7-E6 [26] in AP, but 6.1% faster in inference. In comparison to models with the same input size, our methods achieve overwhelming advantages in terms of accuracy. Real-time is achieved despite a significant increase in time complexity. For instance, CenterNet [73] shows the lowest space and time complexity of 14.43M and 19.34G, respectively, but it achieves only 55.9% accuracy. This mainly stems from the idea that CenterNet [73] employs keypoint triples for

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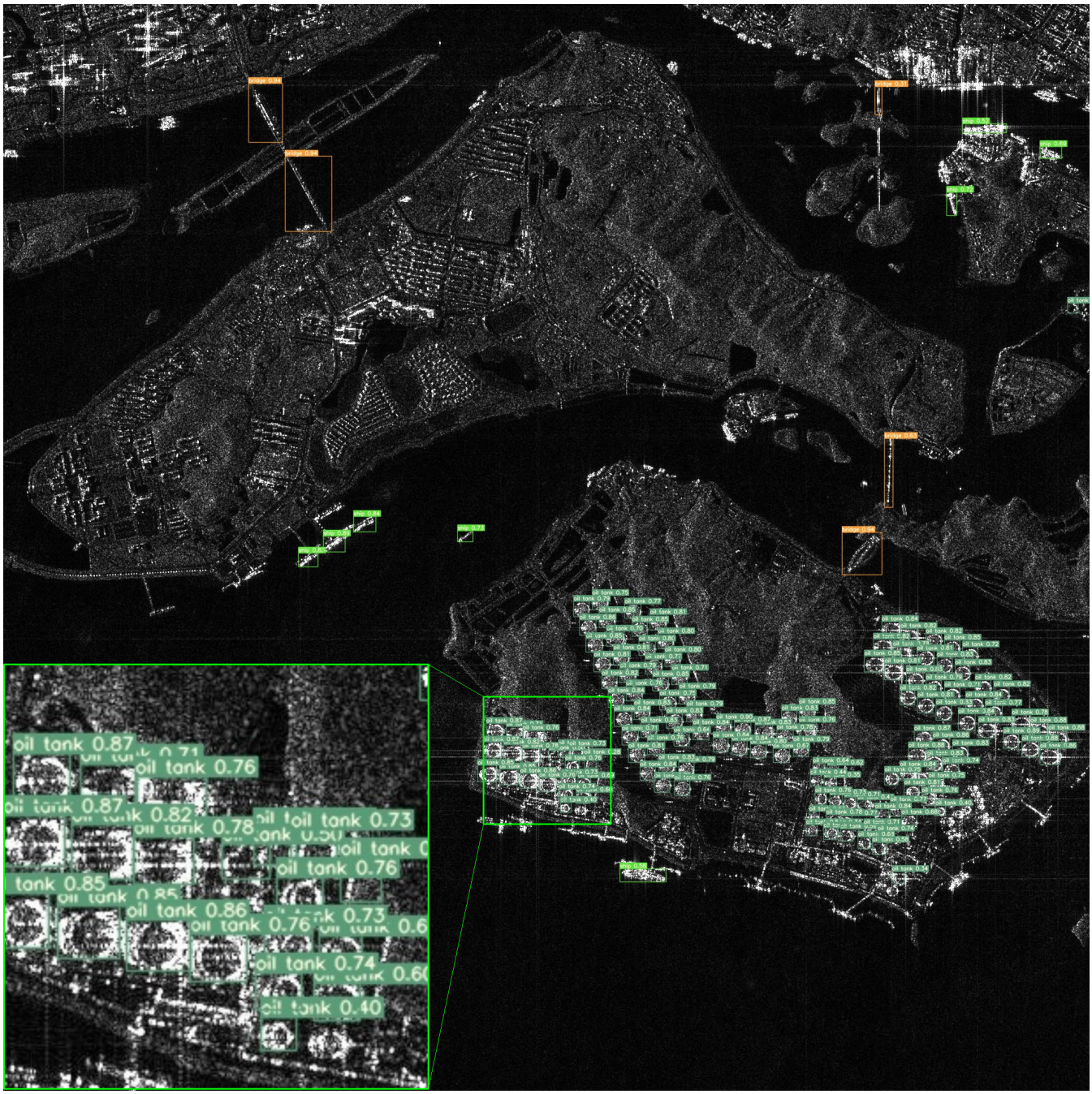


Fig. 11. Multicategory target detection results on SAR images of 2048 × 2048 pixels in MSAR-1.0. Different colors indicate different categories.

target detection, which dramatically reduces time complexity and space complexity while also leading to limited representation ability, especially in complex scenarios. Although DAFDet-YOLOv7-X/100 exhibits the highest space and time complexity of 120.13M and 276.7G, respectively, it achieves an accuracy of 70% and a detection efficiency that fully satisfies the ultralow-latency requirements. There is a direct 14.1% improvement in accuracy. YOLOv7 [26] demonstrates excellent inference speed due to its efficient network structure design, but its accuracy is generally lower than that of DAFDet. It is clear that our methods obtain better results, realize a new set of state-of-the-art speed-accuracy tradeoffs, and that these tradeoffs can be obtained with only one detector, rather than multiple independent detectors. Fig. 10(c) further illustrates the above results. The qualitative results in the first three columns reveal that our method still exhibits excellent target detection performance for scenes with natural clutter interference, strong speckle noise, and heavy occlusion. The qualitative results in the fourth and fifth columns demonstrate the excellent localization and classification capabilities of our model.

b) Challenging SAR-specific detection algorithms: In addition to optical data-based universal detection algorithms, researchers have developed SAR-specific detection algorithms on the basis of universal algorithms or from scratch by considering the characteristics of SAR images.

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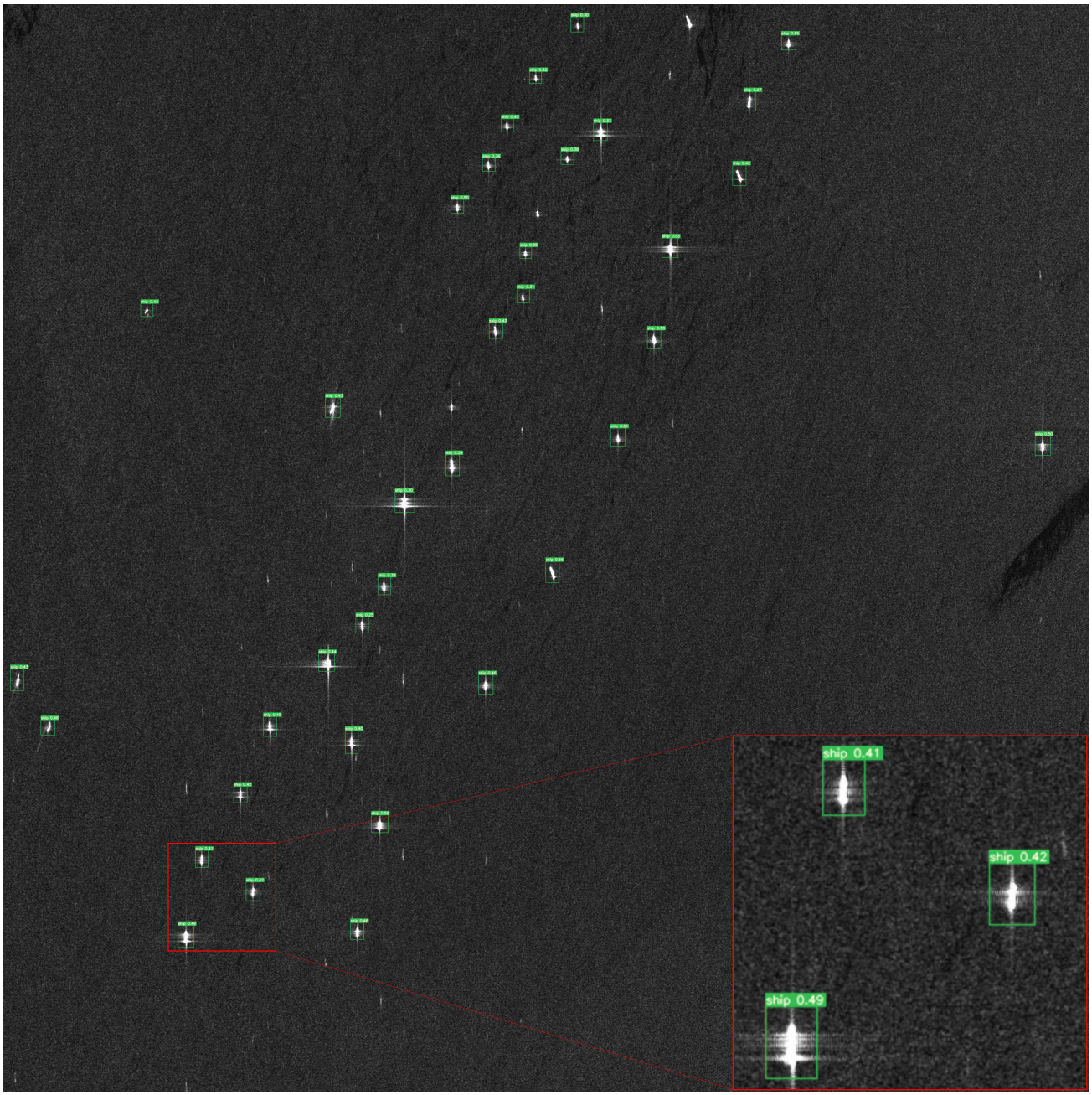


Fig. 12. Tiny target detection results on SAR images of 3072 × 3072 pixels in LS-SSDD-v1.0.

A comprehensive comparison study with the current most advanced SAR-specific detection algorithms is first carried out on SSDD, as illustrated in Table IX. The experimental results indicate that our method achieves the best precision and mAP of 0.969 and 0.981, respectively. AFSar [15] is a highly competitive effort which obtains the optimal recall and $F1$ score of 0.982 and 0.961, respectively. This is made possible by the designed attention enhancement path aggregation network module that highlights SAR target features and suppresses background information. However, it is still 0.004 lower than DAFDet-YOLOv7/100 in mAP, which fully demonstrates the high accuracy of the proposed method.

Next, a comparative analysis is conducted on SAR-Ship-Dataset to examine the speed-accuracy tradeoffs of the algorithms, as presented in Table X. The results reveal that our method attains the best speed-accuracy tradeoff with 96.8% mAP and 9.6 ms inference time. The literatures [22], [23], and [24] all exhibit slower inference speeds of 31.7, 71.2, and 98.3 ms, respectively, whereas SSE-CenterNet [11], ADERLNet-CW [25], and DiffDet4SAR [74] strike a better balance. Obviously, our method exhibits significant advantages with respect to SAR-specific detection methods and achieves new state-of-the-art performance.

2) Comparison With Traditional Methods: Except for optical data-based universal algorithms and SAR-specific detection methods, we also conduct a comparative analysis with representative traditional methods on SSDD, as displayed in Table XI. Due to the difference in fundamental principles, the performance of traditional methods is evaluated in a different way from that of DL-based algorithms, thus we count their

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$F1$ score, precision, and recall simultaneously to enable a fair comparison. The results show that our proposed DAFDet-YOLOv7/100 obtains the optimal $F1$ score and precision with 0.958 and 0.969, respectively, and is much better than OS-CFAR [6], CA-CFAR [5], CHA-CFAR [78], and ASL-CFAR [79] in terms of precision. The CFAR-based methods all result in a higher recall, with OS-CFAR [6] achieving the best with 0.988, which is 0.041 higher than our method. This is mainly due to the presence of a large number of small targets in SSDD. DL-based algorithms usually adopt down-sampling to reduce data storage and computation, and accelerate the convergence of the model, but it will make it difficult to effectively extract and retain the features of small targets, reducing the localization accuracy of the model and thus resulting in lots of missing alarms. Whereas CFAR-based algorithms are generally based on the idea of background statistics, there is no loss of significant features related to small targets, resulting in higher recall.

D. Target Detection on Large-Scale SAR Images

Large-scale SAR image is commonly utilized for practical engineering applications such as environmental monitoring, agricultural yield estimation, and disaster warning. To validate the effect of the developed DAFDet on large-scale SAR images, the SAR images of 2048 × 2048 pixels in MSAR-1.0 are first detected, as depicted in Fig. 11. As can be seen from the detection results, our method still exhibits excellent detection performance. Not only does it provide accurate localization of densely packed targets, but it also allows for high classification confidence and is virtually unaffected by complex backgrounds and strong speckle noise. In addition, there are almost no false alarms (FP = 1), except for a few missing alarms (FN = 4). The inference time is even as low as 12.81 ms, demonstrating the high efficiency of our DAFDet on large-scale SAR images.

Second, the experiment is carried out on SAR images of 24 000 × 16 000 pixels in LS-SSDD-v1.0. Due to memory limitation, these large-scale high-resolution SAR images are cropped into blocks of 3072 × 3072 pixels for detection, as indicated in Fig. 12. The results indicate that our method provides excellent detection performance for tiny targets in large-scale high-resolution SAR images, with only three missing alarms (FN = 3) and no false alarms (FP = 0). Furthermore, the detection results are not affected by speckle noise, demonstrating the ability of our model to effectively disentangle the background noise. The inference time is 13.02 ms, achieving efficient target detection on large-scale high-resolution SAR images.

V. Conclusion

In this article, we propose a unified dynamic architecture with asymptotic fusion enhancement and feature encoding decoupling, which can achieve high-accurate and low-latency SAR target detection in multiple scenarios. First, a dynamic architecture is constructed by cascading two detectors and integrating a designed decision maker. With the adjustment of the threshold in the decision maker, this dynamic architecture can meet the delay requirements of multiple scenarios. Second, AFEFPN is developed to improve the capability of the model to extract effective target information and suppress background noise in complex scenarios. Finally, TODH is proposed to further enhance the ability of the model to classify and localize in complex scenes. The experimental results indicate that DAFDet obtains APs of 70%, 82.2%, and 68.3% on the three publicly available datasets SAR-Ship-Dataset, MSAR-1.0, and SSDD, respectively. Meanwhile, it also obtains excellent detection results on large-scale images in the MSAR-1.0 and LS-SSDD-v1.0 datasets. However, the current architecture is still bloated and the design of the decision maker only considers the multiscale information of targets in SAR images. In the future, we will further optimize this dynamic architecture and the design of its decision maker to make it more robust. In addition, we will focus on multimodal fusion with optical and infrared images to further improve the detection performance.

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